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Full length article

An uncertainty-aware hybrid CNN-transformer ensemble for multiclass brain tumor classification from MRI

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ABSTRACT

Background: Accurate and reliable multiclass brain tumor classification from magnetic resonance imaging (MRI) remains challenging because of substantial variation in tumor morphology, overlapping imaging characteristics, and uncertainty in model predictions. This study aimed to develop and evaluate an uncertainty-aware hybrid CNN–Transformer framework that integrates complementary local and global image representations while incorporating leakage-aware evaluation, calibration analysis, and cross-dataset testing.

Methods: A hybrid architecture combining EfficientNetV2-B0 and Vision Transformer (ViT) representations was developed using cross-modal attention and adaptive feature-level fusion. Predictive uncertainty was assessed using Monte Carlo (MC) Dropout, and probability calibration was evaluated using Expected Calibration Error (ECE) and negative log-likelihood. The primary Kaggle/Masoud brain tumor MRI dataset was screened for exact duplicates and near-duplicate images using cryptographic hashing and perceptual-similarity-based grouping. Because explicit patient identifiers were unavailable, these similarity-derived groups were used to separate related-looking images across data partitions; this procedure does not establish confirmed patient-level independence. Performance was assessed using discrimination, classification, calibration, and uncertainty metrics, with additional evaluation on public brain MRI datasets where applicable.

Findings: On the primary test set, the proposed Hybrid Ensemble achieved 98.10% accuracy (95% CI, 97.37–98.76), with an F1-score of 0.9810 and macro-AUC of 0.9988. It outperformed ViT (97.46% accuracy) and EfficientNetV2-B0 (96.76% accuracy), as well as conventional output-level ensembles (97.08–97.86%). Across ten group-wise folds, the Hybrid Ensemble achieved a mean F1-score of 98.19% (SD, 0.16%) and significantly outperformed EfficientNetV2-B0 (Wilcoxon signed-rank test, $p = 6.03 \times 10^{-11}$) and ViT ($p = 1.22 \times 10^{-6}$). On BraTS 2021, accuracy was 97.34% and F1-score was 0.9673. MC Dropout reduced ECE from 0.031 to 0.017. Correct predictions showed higher confidence and lower predictive entropy than misclassified predictions (maximum softmax confidence, 0.983 vs 0.785; entropy, 0.058 vs 0.536; Kolmogorov–Smirnov statistic, 0.803; $p < 0.001$).

Interpretation: The proposed hybrid framework combines adaptive CNN–Transformer feature fusion with uncertainty-aware inference and demonstrated high discriminative performance, improved calibration, and consistent performance across public datasets. However, these findings represent methodological validation on public 2D MRI datasets rather than evidence of clinical effectiveness. In particular, the absence of confirmed patient identifiers in the primary dataset limits certainty regarding patient-independent generalization. Prospective, multi-institutional validation using standardized clinical MRI data and expert-annotated localization is required before clinical deployment.

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1. Introduction

The brain tumors are one of the deadliest neurological diseases, and early and accurate diagnosis is necessary to increase patient survivals and treatment planning process. The high soft tissue contrast and non-invasive nature of MRI has made it the most commonly used technique for brain tumor detection. Nevertheless, visual diagnosis is time-consuming, subjective, and with a high degree of inter-observer variability which emphasizes the need for automatic classification systems.

Due to their ability of extracting hierarchical features automatically, Convolutional Neural Networks (CNNs) have been widely used in medical image analysis. Recently proposed hybrid CNN models using multiple combinations of CNN backbones and classic classifiers have shown robust multimodal brain tumor classification with computational efficiency.[1] Lighter weight CNN models and transferring learning models also improve performance while reducing computational demands yet even to a large enough degree to be viable in clinical practice.[2] The attention-augmented lightweight CNNs have also optimized the feature representation, to better support interpretability and efficiency.[3] Intuitively, it is shown that the advantage of CNN models like feature extraction in close vicinity of one block while they do not properly learn information capture from long-range and global contexts which motivates the fusion with transformer encoder.[4]

Apart from CNNs, attention-based transformers have been a strong contender due to their capability to capture global contextual dependencies. Hybrid architectures based on CNN backbones equipped with self-attention blocks, recurrent units or transformer modules have achieved competitive results by learning local and global tumor characteristics simultaneously,[5,6] Attention-augmented hybrid architectures, like T-FSPANNet, fuse multi-scale feature representation with pyramidal attention mechanisms and are proven to reach a highly competitive accuracy of more than 99.8%.[2] Another ensemble hybrid pipeline such as BRAIN-META integrates CNNs and Vision Transformers in a meta-learning context to increase the robustness and interpretability of the predictions.[6] The Vision Transformer is reported outperforming the traditional CNNs in various multiclass MRI classification tasks by systematical reviews, which reflects the trend of mixed and attention-derived architectures rising among medical imaging.[7]

Despite these achievements, to the best of our knowledge, the majority of current studies still use individual deep learning models, static hybrid architectures or traditional ensemble approaches which may yield suboptimal generalization and robustness in heterogeneous or imbalanced data. Score-level fusion and meta-ensemble strategies enhance prediction performance at the expense of dynamic feature fusion or predictive uncertainty quantification, thus hindering them from applying as a practically offline decision support in a clinical setting.[6,7] As a result, there is still a gap for adaptive hybrid systems that can integrate both CNN and transformer representations, meanwhile taking reliability-aware inference into consideration.

The proposed architecture consists of a fine-tuned Vision Transformer captures global contextual dependencies and an enhanced EfficientNetV2-B0 backbone feature by attention-based feature refinement for discriminative local patterns extraction. In contrast to the traditional hybrid strategy based on static feature fusion, we introduce adaptive feature-level fusion for dynamic weighting of convolutional and transformer representations in light of different tumor patterns. Furthermore, predictive uncertainty is estimated by Monte Carlo Dropout–based inference to detect low-confidence predictions and for reliability-oriented assessment.

New to architectures, we focus on transparency of the experimental methodology and reproducibility via controlled reporting about data - provenance, leakage-aware splitting strategies as well as cross-dataset validation. The key contributions of this paper can be summarized as follows:

- Development of an adaptive hybrid CNN–Transformer architecture that jointly captures attention-refined local representations and global contextual dependencies.
- Introduction of a dynamic feature-level fusion mechanism that balances convolutional and transformer representations instead of employing fixed late-fusion strategies.
- Integration of uncertainty-aware inference using Monte Carlo Dropout to enable reliability assessment and calibration analysis.
- Comprehensive benchmarking and ablation studies demonstrating the effectiveness of the proposed reliability-aware hybrid design.

Together, these contributions position the proposed framework as a methodologically robust advancement rather than a performance-benchmarking study alone.

2. Related work and motivation

Multiclass brain tumor classification using magnetic resonance imaging (MRI) remains a challenging problem in medical image analysis because of substantial intraclass heterogeneity, interclass visual similarities, and variability in tumor size, anatomical location, and image acquisition protocols across clinical centers. These factors can complicate the differentiation of glioma, meningioma, pituitary tumors, and non-tumorous brain images, particularly when models are evaluated using heterogeneous public datasets. Consequently, the development of reliable classification systems requires not only effective feature extraction but also appropriate architectural integration, robust evaluation, and careful assessment of predictive uncertainty.

2.1. Deep learning architectures for brain tumor classification

Convolutional neural networks (CNNs) have been widely investigated for brain tumor classification because their hierarchical feature representations capture local patterns, including edges, textures, and spatially localized structural characteristics. Recent studies have explored both lightweight CNN architectures and transfer-learning approaches to improve predictive performance while limiting computational requirements. EfficientNet-based models, including combinations of EfficientNet-B0 and lightweight Vision Transformers (ViTs), have demonstrated the potential of integrating computationally efficient convolutional features with attention-based representations.[8] Other studies have investigated pretrained ResNet variants and EfficientNet- or Swin Transformer-based approaches, reporting promising classification performance on their respective evaluation datasets.[1,9] These findings support the continued relevance of CNNs while highlighting the potential benefits of incorporating broader contextual information.

Vision Transformers have extended this approach by representing images as sequences of patches and modeling relationships between spatially separated regions through self-attention. Their application to brain tumor classification has produced promising results, including studies reporting high accuracy for multiclass classification and glioma grading.[10,11] A systematic review of Vision Transformers in medical imaging has also examined their potential advantages over conventional CNNs, while emphasizing the importance of architecture selection, pretraining, and task-specific evaluation.[7] Nevertheless, performance differences across studies should be interpreted in the context of dataset composition, class definitions, preprocessing, and evaluation protocols rather than attributed exclusively to the underlying architecture.

Hybrid CNN–Transformer architectures seek to combine the complementary strengths of local convolutional feature extraction and global contextual modeling. Recent investigations have examined lightweight CNN–ViT combinations, attention-guided feature integration, and hierarchical transformer architectures for brain tumor classification. T-FSPANNet, for example, incorporates tri-attribute and pyramidal attention-based feature fusion, whereas other studies have investigated Swin Transformer-based approaches and hybrid

architectures to improve representation learning and computational efficiency.[12] Task-specific comparisons have further demonstrated that the relative performance of CNNs and transformer-based models can vary according to the imaging task and experimental design.[13] Related approaches, including SEMDNet and ShallowMRI, have also contributed to the development of deep-learning methods for brain imaging, although their specific tasks and evaluation settings must be distinguished when comparing classification with segmentation outcomes.[14,15]

More recent research continues to investigate alternative methods of combining convolutional and attention-based representations. The BRAIN-META CNN–ViT meta-ensemble framework and the H-ConvNeXt–Swin architecture illustrate complementary directions in hybrid model development, while the reported T-FSPANNet framework emphasizes attention-guided multiscale feature fusion.[16] These approaches demonstrate the methodological diversity of contemporary brain tumor classification research. However, their reported accuracy values cannot be compared reliably without considering the datasets, class distributions, partitioning strategies, and evaluation protocols used in each study. High accuracy on a particular benchmark does not, by itself, establish superior generalization to independent patients or external clinical populations.

2.2. Predictive uncertainty and probability calibration

Reliable biomedical AI requires consideration of predictive uncertainty in addition to classification performance. A model may assign a high probability to an incorrect prediction, meaning that strong discrimination does not necessarily imply well-calibrated confidence. Bayesian neural networks, Monte Carlo (MC) Dropout, deep ensembles, and evidential learning have been investigated as approaches to estimating predictive uncertainty in medical imaging.[17] Other studies have combined uncertainty estimation with feature fusion and attention mechanisms, including DFuse-Net for multimodal brain tumor segmentation,[18] ensemble-based uncertainty-aware segmentation on BRISC,[19] and Bayesian or attention-based approaches to medical-image classification and anomaly detection.[20,21]

These studies provide methodological context for uncertainty-aware modeling, but their findings must be interpreted according to the specific tasks investigated. Segmentation uncertainty, anomaly-detection performance, and multiclass classification confidence are related but distinct research problems. In particular, improvements in segmentation overlap metrics, such as the Dice coefficient, cannot be directly interpreted as evidence of improved multiclass classification accuracy or calibration.

Several complementary methods can be used to assess uncertainty and confidence. MC Dropout estimates predictive variability through repeated stochastic forward passes, while deep ensembles examine variability across independently trained models. Temperature scaling provides a comparatively simple post-hoc calibration approach by adjusting the scale of predicted logits. Guo et al. established an influential methodological foundation for evaluating and improving the calibration of modern neural networks.[22] Together, these approaches address different aspects of model reliability: classification metrics quantify predictive performance, calibration metrics assess the agreement between confidence and observed correctness, and uncertainty analyses examine the model's predictive variability or confidence in individual cases.

Accordingly, accuracy and F1-score alone provide an incomplete assessment of a medical-image classifier. Calibration measures, including expected calibration error (ECE) and negative log-likelihood (NLL), can complement discrimination metrics by characterizing different aspects of predicted probabilities. Their interpretation depends on the precise implementation, including the calibration procedure, binning strategy, and evaluation sample. A rigorous evaluation should therefore distinguish improvements in classification performance from improvements in calibration and should avoid treating these outcomes as interchangeable evidence of reliability.

2.3. Explainability, multiscale representation, and clinical relevance

Interpretability and explainability have become important considerations in the development of biomedical AI systems. Transformer-based medical-image studies have investigated attention patterns and feature representations to improve understanding of model behavior in decision-support applications. Hierarchical attention mechanisms have also been explored to integrate fine-grained spatial features with broader contextual representations in complex medical images. These approaches are relevant to brain MRI, where diagnostically informative features may occur at different spatial scales and where both local anatomical patterns and relationships between image regions may contribute to classification.

Related research has examined attention-guided feature selection, explainable deep-learning architectures, and hybrid representation learning for medical-image analysis. Broader reviews of deep learning in cancer and biomedical imaging have emphasized the importance of evaluating predictive performance alongside computational efficiency, interpretability, and potential clinical applicability. Research on explainable AI and transfer learning further highlights the need to examine how models use learned image representations when developing clinically oriented applications.

However, attention maps and post-hoc explanation methods should not automatically be interpreted as evidence of clinical validity. Their usefulness depends on the explanation method, the stability of the resulting outputs, and the extent to which the highlighted image regions support a defensible interpretation of model behavior. Similarly, high classification performance does not establish that a model is suitable for clinical deployment. External validation, assessment of failure cases, and transparent reporting remain necessary to characterize the strengths and limitations of an automated system.

2.4. Reproducibility and evaluation of medical-imaging AI

Evaluation design is particularly important when public datasets aggregate images from multiple sources. Duplicate and near-duplicate images may affect estimates of performance if related images are distributed across training, validation, and test partitions. Exact-duplicate removal and perceptual-similarity-based grouping can reduce the risk of duplicate-related contamination. Nevertheless, image similarity is not a reliable substitute for verified patient identifiers: images from the same patient may appear dissimilar, whereas images from different patients may appear similar.

Cross-dataset evaluation provides additional evidence about performance under changes in image sources, acquisition conditions, and dataset composition. However, it does not automatically establish clinical effectiveness or patient-independent generalization. Transparent reporting of data provenance, preprocessing, partitioning, model selection, metric definitions, and uncertainty estimation is therefore essential for interpreting results and enabling reproducibility. The Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 Update provides relevant reporting guidance for medical-imaging AI research.[23]

Recent hybrid architectures, including the H-ConvNeXt–Swin framework, illustrate continued progress in the design of brain tumor classification systems.[16] However, comparisons between such architectures and other CNN–Transformer methods should account for differences in datasets, training procedures, and evaluation protocols. The need for reproducible, appropriately qualified evaluation is especially important when studies report small differences in accuracy or make claims about model superiority.

2.5. Research gap and motivation for the present study

Despite progress in CNNs, Vision Transformers, hybrid architectures, and uncertainty-aware medical imaging, several methodological challenges remain relevant to multiclass brain tumor classification. First, conventional output-level ensembles combine model predictions but do not necessarily learn an adaptive interaction between convolutional and transformer-based feature representations. Second, strong classification performance does not guarantee well-calibrated probabilities or appropriately characterized uncertainty. Third, performance comparisons can be difficult to interpret when evaluation protocols, data partitioning, and metric definitions are inconsistently reported.

The present study investigates a hybrid architecture that integrates EfficientNetV2-B0 and Vision Transformer representations through cross-model attention and adaptive feature-level fusion. EfficientNetV2-B0 is intended to capture local spatial and textural characteristics, while ViT provides contextual representations of relationships between image patches. The proposed fusion mechanism is designed to integrate these complementary representations within a unified classification framework rather than relying exclusively on static output-level aggregation.

The study also examines predictive uncertainty using MC Dropout and evaluates probability calibration using ECE and NLL. Conventional ensemble methods are included as comparators, and ablation experiments are intended to assess the contributions of the model components. Where supported by the experimental design, cross-dataset evaluation provides additional evidence regarding performance on data from a different source. The primary Kaggle/Masoud dataset is partitioned using similarity-derived image groups to reduce the risk of duplicate-related contamination; because explicit patient identifiers are unavailable, this procedure does not establish confirmed patient-level independence.

The intended contribution is therefore methodological rather than a claim of universally superior accuracy: to evaluate adaptive CNN–Transformer feature fusion alongside uncertainty estimation, calibration analysis, component ablation, and appropriately qualified dataset-level evaluation. The findings should be interpreted according to the verified experimental results and the limitations of the available datasets. Table 1 summarizes representative studies, their methodological approaches, and the remaining challenges that motivate the present investigation.

3. Methodology

The novel system for tumor classification of brain, illustrated in Fig. 1 has an organized nature. It starts with Data Collection, then performs Data Preprocessing, Data Augmentation and Data Annotation as prerequisites for Model Training. Two deep learning models, EfficientNetV2 B0 and Vision Transformer are utilized to perform feature extraction as well as preserve long-range relationships. These models are united in an Ensemble Learning engine, accompanied by methods namely Hybrid Ensemble, Weighted Average Ensemble and Geometric Mean Ensemble to enhance the classification accuracy. The last phase is the Classification stage that tests the model performance for tumor detection.

3.1. Dataset description

For multi-class classification of brain tumors, we used the Brain Tumor MRI Dataset for training and evaluation.[11] The dataset comprises 7023 contrast-enhanced brain MRI slices aggregated from Figshare, SARTAJ, and Br35H sources. The images are categorized into four classes: glioma (1621 samples), meningioma (1645 samples), pituitary tumor (1757 samples), and no tumor (2000 samples).[25]

Because the Kaggle/Masoud dataset does not provide explicit patient identifiers, true patient-level partitioning could not be verified. We therefore applied a similarity-based grouping procedure to reduce the possibility that identical or near-duplicate images would be distributed across the training, validation, and test sets. First, exact duplicate images

Table 1
Recent State-of-the-Art Studies on Multi-Class Brain Tumor MRI Classification, Their Architectures, Datasets, Performance, and Limitations.

References	Year	Architecture	Dataset	Key Findings	Limitations
17	2024–2025	Bayesian Neural Networks, Monte Carlo Dropout, Deep Ensembles, Evidential Learning	BraTS	Improved calibration, reduced false positives	Mostly segmentation; limited hybrid CNN–Transformer evaluation
18	2025	Disentangled fusion + uncertainty estimation	BraTS	Outperformed SOTA in multi-modal tumor segmentation	Focused on segmentation, not multi-class classification
19	2025	Ensemble + uncertainty modeling	BRISC	Dice score 0.798	Limited to segmentation; no transformer integration
20	2025	Attention + Bayesian networks	Multi-dataset	Improved accuracy and calibration	Limited application to hybrid CNN–Transformer models
21	2025	VAE + multi-head attention	BraTS	ROC-AUC 0.83, PR-AUC 0.83	Focused on anomaly detection; not classification
8	2024	EfficientNet B0 + Tiny ViT 5 M	Msoud Nickparvar Kaggle – Multi-class MRI	Accuracy 98.41%; efficient computation	Limited interpretability, single hybrid combination
12	2025	T-FSPANNet (T Block + FSPAN + RGAP)	Msoud Nickparvar Kaggle – Multi-class MRI	Accuracy 99.89%; improved interpretability	Complexity; specific to T-FSPANNet design
13	2025	Swin-based Transformers	Msoud Nickparvar Kaggle – Multi-class MRI	Accuracy 99.37%; optimized computational cost	Single transformer variant; generalization not tested
9	2025	CNN–ViT hybrid	Msoud Nickparvar Kaggle – Multi-class MRI	Highlighted architecture-specific performance variations	No uncertainty modeling
24	2024	ResNet pretrained models	Msoud Nickparvar Kaggle – Multi-class MRI	Accuracy up to 99.2%; high AUC & sensitivity	CNNs lack global context modeling
10	2024	EfficientNet + Swin ViT	Msoud Nickparvar Kaggle – Multi-class MRI	Accuracy up to 98.72%; improved hybrid performance	Deterministic predictions; no uncertainty estimation
14	2024–2025	ViT, CNN–Transformer hybrids	Glioma & Msoud Nickparvar Kaggle – Multi-class MRI	Accuracy 99.21% glioma grading, 98.66% multi-class	Most studies do not model predictive uncertainty
15	2024–2025	Unified & lightweight frameworks	Benchmark datasets	Strong segmentation and classification performance	Limited hybrid feature-level fusion; reproducibility not addressed
11	2024	CNN vs ViT	Multiple datasets	Highlights superiority of ViTs with pretraining	Limited to review; no new framework proposed

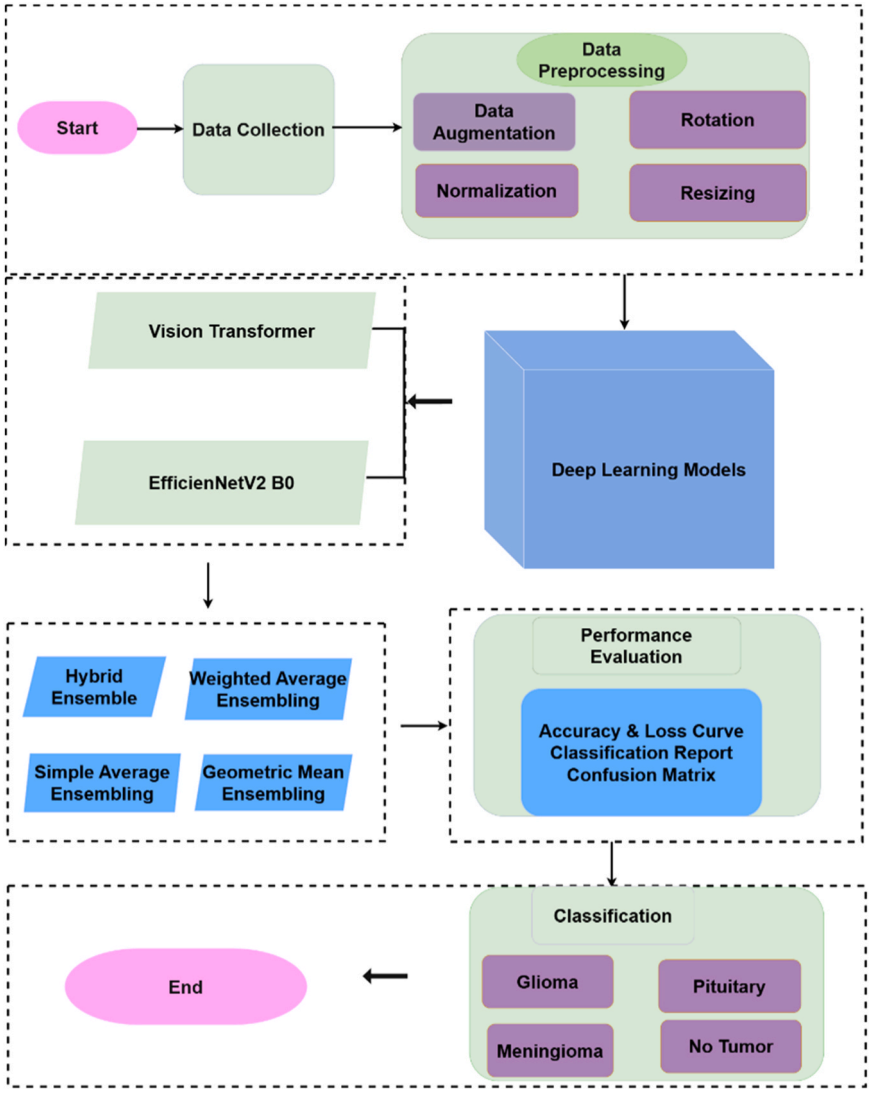


Fig. 1. Proposed block diagram for Brain Tumor Classification.

were identified using MD5 hash comparison. Perceptual hashing was subsequently used to identify visually similar images and construct similarity-derived groups. These groups represent image-similarity clusters rather than confirmed patients or clinical imaging sessions. The groups were assigned exclusively to individual dataset partitions to reduce the risk of contamination from duplicated or highly similar images.

The dataset was divided into training (5107 images), validation (549 images), and test (1367 images) sets. Partitioning was performed at the similarity-cluster level using a fixed random seed to support reproducibility. The numerical group counts reported in Table 2 represent perceptual-similarity clusters and must not be interpreted as counts of confirmed unique patients. The resulting partitioning was intended to reduce the risk of cross-set contamination associated with duplicate and near-duplicate images.

Although similarity-based grouping can reduce the risk of duplicate-related contamination, it does not establish patient-level independence. Images from the same patient may appear dissimilar and remain in different partitions, whereas images from different patients may appear sufficiently similar to be assigned to the same cluster. Consequently, residual subject-level dependence and source-dataset bias cannot be excluded, and the resulting evaluation should not be described as definitively patient-independent or leakage-free.

The reproducibility and interpretation of perceptual-hash grouping depend on the hash algorithm, hash length, distance metric, similarity threshold, clustering procedure, and treatment of transitive matches. These implementation details, together with the numbers of exact duplicates removed and images retained in each similarity cluster, should

Table 2
Similarity-Group Distribution Across Dataset Splits for the Kaggle/Masoud Dataset.

Split	Images	Similarity Groups	Average Images per Similarity Group	Near-Duplicate Removed	Train→Val Shared Patients	Train→Test Shared Patients	Val→Test Shared Patients
Train	5107	3169	1.61	2134	0	0	0
Validation	549	357	1.54	0	0	0	0
Test	1367	1299	1.05	0	0	0	0
	7023	4825	-	2134	-	-	-

be reported where available. Where explicit patient identifiers are available, patient-level grouping should take precedence over image-similarity-based proxies.

The overall distribution of images across the four tumor categories is summarized in Table 3, while the class-wise distribution across the training, validation, and test sets is presented in Table 4. For the Kaggle/Masoud dataset, the reported group counts refer to similarity-derived image clusters rather than verified patient counts.

The Masoud Nickparvar brain tumor MRI dataset is a publicly available benchmark dataset used in brain tumor classification research.[26–29] Although the dataset aggregates images from multiple sources and may contain labeling inconsistencies, its use in previous studies supports its utility as a methodological benchmark for comparative evaluation. Nevertheless, the heterogeneous source composition and absence of explicit patient identifiers impose limitations on the interpretation of model performance and its generalizability to independent patients and clinical settings.

The primary dataset contains four image-level classes with sample counts of 1621 for glioma, 1645 for meningioma, 1757 for pituitary tumor, and 2000 for no tumor. Although the class frequencies are not identical, the distribution is not severely imbalanced at the image level. Stratified partitioning was used to preserve class proportions as closely as permitted by the similarity-group constraints. The Vision Transformer branch was trained using focal loss with label smoothing as described in Section 3.4. The loss function used for the EfficientNetV2-B0 branch and the final jointly trained hybrid model should be reported separately according to the actual implementation. Performance should be presented using per-class precision, recall, F1-score, and support, alongside macro- and weighted-average measures, so that aggregate accuracy does not obscure class-specific errors.

In addition, the model was evaluated on the BraTS 2021 benchmark dataset to assess cross-dataset generalization. Any claim of patient-level separation in the BraTS 2021 experiments must be supported by the dataset's subject identifiers and the documented partitioning procedure. Where subject identifiers were used to assign all images from each subject exclusively to one partition, this should be reported as subject-level grouping. Perceptual-similarity clustering alone should not be described as guaranteeing zero subject overlap.

3.2. Data preprocessing and augmentation

Standardizing the height and width of the MRI pictures in the dataset is crucial for best results because their dimensions vary.[25,30] All MRI images were resized to 224 × 224 pixels and intensity-normalized to standardize input distributions. Background masking was applied to remove irrelevant regions such as the skull and other non-brain structures. Brain regions were extracted using a region-of-interest (ROI) mask to ensure that model predictions were based on tumor regions rather than unrelated anatomy. The preprocessing pipeline was applied before model input while preserving the original diagnostic content. Brain/-background masking was applied consistently across the dataset, followed by intensity normalization. During model development, augmentation parameters were restricted to transformations considered unlikely to alter the underlying diagnostic class. No stochastic augmentation was applied to validation or test images.

Table 3
Distribution of Training, Testing and Validation Samples across Tumor Categories (All reported results were obtained using Kaggle P100 GPU infrastructure.).

Tumor Category	Train	Test Set Count	Validation	Total
Glioma	1134	352	135	1621
Meningioma	1156	361	128	1645
No Tumor	1509	325	166	2000
Pituitary	1308	329	120	1757
Overall Count	5107	1367	549	7023

Table 4
Class-wise Breakdown Across Similarity Groups and Dataset Splits.

Tumor Class	Train Images	Train Patients	Val Images	Val Patients	Test Images	Test Patients
Glioma	1127	908	135	114	359	362
Meningioma	1147	931	128	117	370	371
No Tumor	1494	482	166	46	340	218
Pituitary	1215	848	120	80	422	343
Total	5107	3169	549	357	1367	1299

The complete preprocessing and augmentation workflow applied to the MRI dataset is illustrated in Fig. 16. The pipeline consists of sequential stages including raw MRI input acquisition, region-of-interest extraction through brain masking, intensity normalization, and controlled augmentation operations applied during training. Each stage is explicitly labeled with representative parameter ranges to ensure transparency regarding the transformations applied to the input data.

To reduce overfitting and improve generalization, geometric (rotation, flipping, cropping) and photometric (brightness, contrast) augmentations were applied only on the training set. Validation and test images were deterministically resized and normalized to prevent evaluation bias.

To ensure fair comparison across model branches, preprocessing and augmentation were matched as closely as possible across frameworks:

For PyTorch-based models (Vision Transformer and hybrid ensemble), augmentations included random resized cropping, horizontal and vertical flipping, limited rotations, and brightness/contrast changes while preserving clinically relevant structures.

For EfficientNetV2-B0 (TensorFlow), a stronger augmentation pipeline was used to compensate for inter-scanner intensity differences, including rotations, spatial shifts, shear transformation, zooming, contrast modulation, reflection padding, and random channel shifts. Parameter ranges were aligned to ensure comparable augmentation effects across frameworks. Example augmentations are illustrated in Fig. 2, while the detailed preprocessing and augmentation pipeline applied to the brain tumor MRI dataset is illustrated in Fig. 16.

3.3. EfficientNetV2-B0 architecture

A CNN backbone EfficientNetV2-B0 was chosen, because it had the best trade-off between accuracy and computational efficiency. The key components of the architecture are shown in Fig. 3. It is based on depthwise separable convolutions and compound scaling for efficient feature extraction, with several architectural enhancements to adapt the model for brain tumor MRI analysis.

The modified architectural components used in this work are illustrated in Fig. 3, which focuses on the customized modules relevant to the proposed framework rather than reproducing the entire canonical EfficientNetV2 architecture.

3.3.1. Squeeze-and-excitation (SE) blocks

Squeeze-and-Excitation (SE) blocks were utilized for channel-wise feature recalibration which are beneficial to the network in focusing on diagnostically important features while ignoring unimportant activations. This mechanism is very useful for finding subtle textural changes related to the tumor region of MRI images.

3.3.2. Dual pooling strategy

We utilized a global average pooling and a global max pooling to capture the holistic context information and highly discriminative local activations. The fused representation also improves the model's capabilities of discriminating tumor from non-tumor regions, particularly in challenging cases with low contrast boundaries.

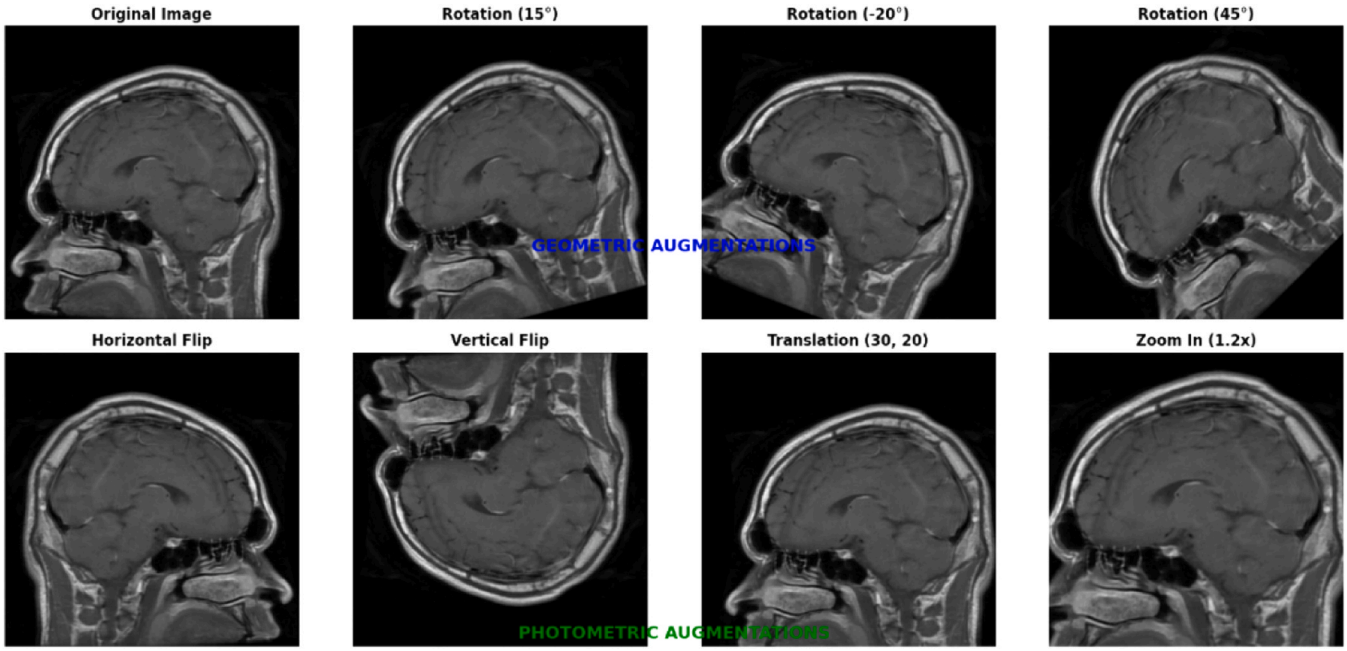


Fig. 2. Data Augmentation.

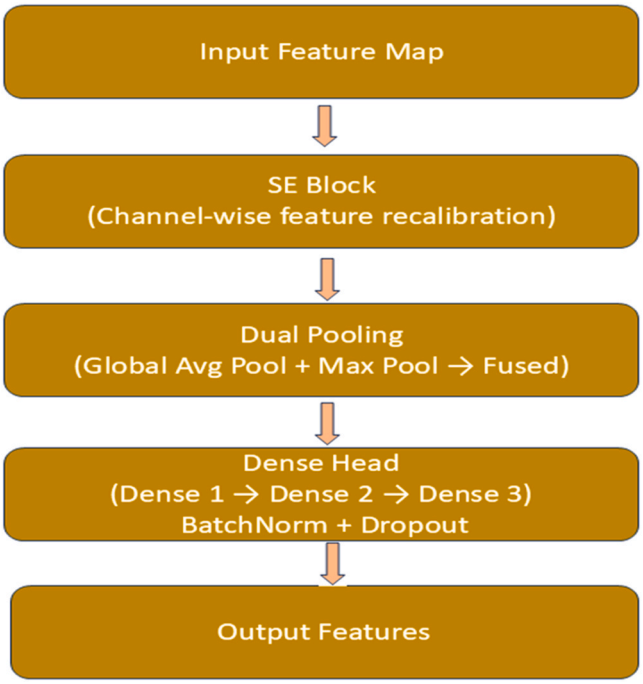


Fig. 3. Key components of EfficientNetV2-B0.

3.3.3. Dense layers and regularization

The original classification head was replaced with a three-stage dense architecture to improve representational capacity. Batch normalization and dropout were applied after each dense layer to stabilize training and reduce overfitting, thereby enhancing generalization across diverse MRI patterns.

3.3.4. Loss function and training strategy

Training was performed with label-smoothed cross-entropy loss to mitigate overconfidence and enhance robustness towards label noise. We first did a warm-up phase before training and used a cosine annealing learning rate schedule for stable convergence. We used the Adam Optimizer for optimization.

3.3.5. Test-time augmentation

Inference stability was further improved through test-time - augmentation by creating several augmented copies of each test image. The final predictions were done by taking average of the softmax probabilities of the augment samples, giving more stability under minimal spatial changes.

3.3.6. Compound scaling and computational efficiency

EfficientNetV2-B0 uses compound scaling to simultaneously scale - network depth, width, and input resolution while demonstrating strong performance at a reduced computational cost. Architectural optimisations allow for running in real-time on low-end hardware, making the model suitable for realistic clinical use.

3.3.7. Model implementation

The EfficientNetV2-B0 model was implemented using TensorFlow and Keras, initialized with ImageNet-pretrained weights and fine-tuned on the brain tumor MRI dataset using GPU acceleration.

3.3.8. Model implementation

We used TensorFlow 2 and Keras for the model. It achieves the Top-1 accuracy of 0.796 and the Top-5 accuracy of 0.953. Activation weights were initialized with those of the pre-trained EfficientNetV2B0 model using ImageNet. The model was optimized for the brain tumor MRI dataset and experiments were executed on a GPU machine to accelerate testing.

3.4. Vision transformer (ViT)

The Vision Transformer (ViT) architecture was employed to capture global contextual dependencies inherent in brain MRI scans. The key components of the ViT model used in this work are illustrated in Fig. 4. Using a patch-based representation, ViT models long-range relationships through self-attention, which is particularly effective for handling irregular tumor boundaries spanning multiple anatomical regions. The architecture components relevant to the proposed hybrid framework are

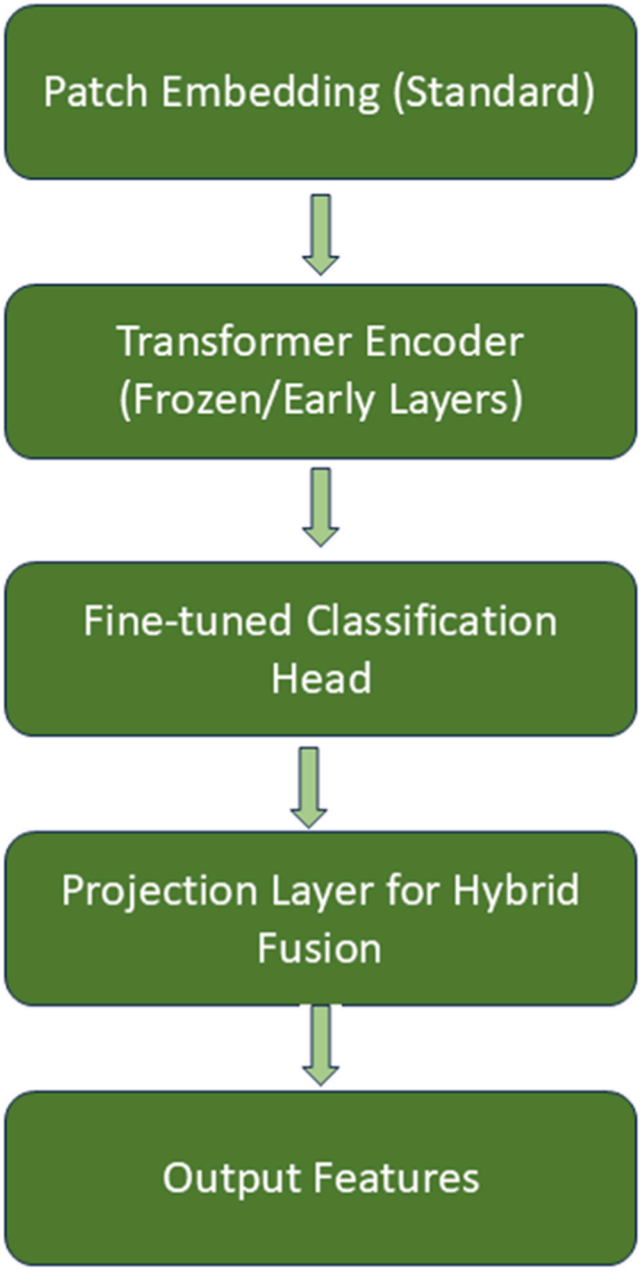


Fig. 4. Key components of Vision Transformer (ViT).

summarized in Fig. 4, highlighting the fine-tuned classification head and projection layers used for feature-level fusion.

3.4.1. Architecture and fine-tuning

Input images were divided into non-overlapping patches and embedded with positional encodings. A transformer encoder processed the token sequence, and the classification head predicted tumor categories. To balance efficiency and performance, early transformer layers were frozen, while later layers and the classifier head were fine-tuned.

3.4.2. Loss design, label smoothing and evaluation

Focal loss combined with label smoothing was employed to address class imbalance and improve generalization. The ViT demonstrated strong performance, particularly in modeling global anatomical context. The formulation of Focal Loss is given by:

$$L_{focal} = - \alpha (1 - p_t)^{\gamma} \log(p_t)$$

where: p_t represents the predicted probability assigned to the correct class, α denotes the balancing coefficient, γ is the focusing term (set to 2).

In addition to this, Label Smoothing is applied as a regularization method to prevent the classifier from becoming excessively confident. By distributing a small portion of the target probability mass across other classes, the model learns smoother decision boundaries. The smoothed loss function is expressed as:

$$L_{smoothing} = - \Sigma_i [(1 - \varepsilon) \cdot 1(y = i) + \varepsilon/K] \log(p_i)$$

where: ε is the smoothing factor (set to 0.1), K denotes the total number of output classes.

3.5. Hybrid ensemble architecture

The proposed Hybrid Ensemble model fuses EfficientNetV2-B0 (CNN) and Vision Transformer (ViT) at the feature representation level rather than at the output level, leveraging the complementary strengths of convolutional and transformer-based models for multiclass brain tumor MRI classification. The overall architecture is illustrated in Fig. 5.

Backbone Features

- EfficientNetV2-B0 (CNN): Generates spatial feature maps of shape (B, 7, 7, 1280) capturing local structural and textural MRI patterns.
- ViT-B/16 (Transformer): Produces token embeddings of shape (B, 196, 768), encoding global contextual information across the image.

Feature Alignment and Fusion

- CNN extracted feature maps, which are further spatially pooled for dimension reduction and spatial correspondence preservation.

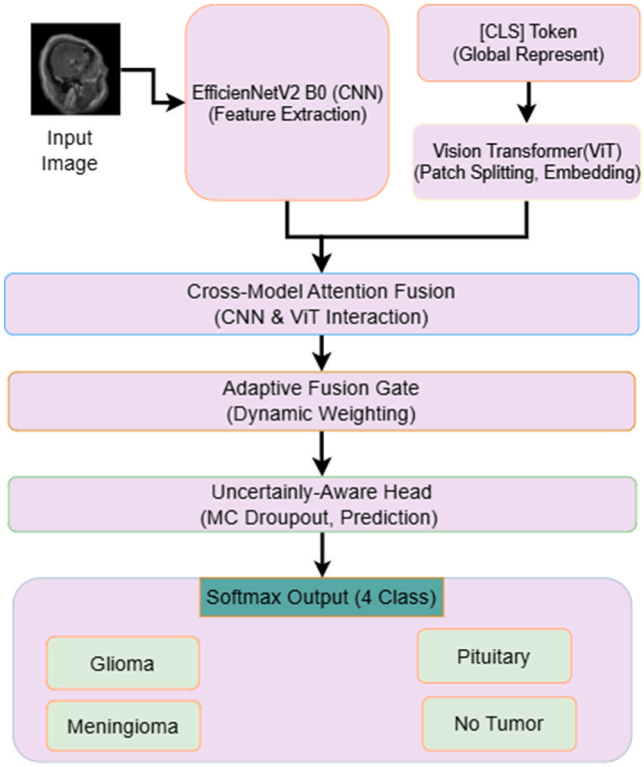


Fig. 5. Architecture of the Hybrid Ensemble showing tensor dimensions, CNN and ViT feature maps, cross-modal attention fusion, and adaptive fusion gate.

- Both CNN and ViT feature are projected into a shared 512-dimensional embedding space with learnable linear layers for meaningful interaction between the modalities.
- Cross-modal attention is employed: CNN and ViT features are projected through learnable projection matrices, attention scores are obtained by dot-product similarity and normalized using softmax. This allows selectively attend informative local regions that are conditioned on global context.

3.5.1. Mathematical formulation of cross-modal attention and adaptive feature fusion

Let $X_C \in R^{B\times NC\times 1280}$ denote the spatially pooled EfficientNetV2-B0 feature representation and $X_V \in R^{B\times NV\times 768}$ denote the ViT token representation, where B is the batch size. The CNN feature map has 7×7 spatial locations, giving (N_C=49), whereas ViT-B/16 produces (N_V=196) patch tokens for a 224×224 input, excluding any class token.

3.5.2. Projection into a shared embedding space

The two representations are projected into a common d= 512-dimensional embedding space:

$C=X_CW_C + b_C$, $V=X_VW_V + b_V$,

where $W_C \in R^{1280\times 512}$ and $W_V \in R^{768\times 512}$

3.5.3. Cross-modal attention

CNN-conditioned attention over transformer tokens is calculated as: Cross-modal attention is then computed by using one modality to query the other. For example, CNN-conditioned transformer attention is defined as:

$Q_C = C\ W_Q$, $K_V = V\ W_K$, $V_V = V\ W'_V$,

$$A_{C\to V} = \text{softmax}\{(Q_C K\frac{T}{V}) \ \Big/ \ \sqrt{d_k}\}$$

$$\widetilde{C} = A_{C\to V}V_V$$

An analogous operation can be performed in the reverse direction to allow transformer representations to condition convolutional representations. The resulting modality-specific representations are aligned in the shared embedding space.

3.5.4. Adaptive fusion gate

Let (c) and (v) denote the resulting dimensionally aligned CNN and transformer vectors. Their concatenation is passed through a learnable gate:

$\alpha = \sigma\ (W_g\ [C; V] + b_g)$

where [C;V] denotes concatenation, W_g and b_g are learnable parameters, and $\sigma\ (\cdot)$ is the sigmoid function. The final fused representation is:

$F = \alpha \odot C + (1-\alpha) \odot V$,

where $\odot$ denotes element-wise multiplication. Thus, α dynamically controls the relative contribution of the CNN and transformer representations for each input.

The gate provides input-dependent feature weighting, while cross-modal attention permits each representation to incorporate information from the other modality. These equations describe the intended operations; the final implementation, including any residual connections, pooling, multi-head attention, normalization, and dropout, must be reported explicitly to ensure that the mathematical description matches the trained model.

3.6. Hyperparameter sensitivity analysis

To evaluate the effect of major hyperparameters on the performance of our method, we have run a systematic ablation study. The hyperparameters evaluated included:

- Learning rate: 1e-3, 1e-4, 5e-5
- Dropout rate: 0.1, 0.3, 0.5
- Fusion gate weight: 0.25, 0.5, 0.75

All configurations were trained and tested on the same data splits, pre-processing and training schedule. We quantified model performance with Accuracy, F1-score, and Expected Calibration Error (ECE). Hyperparameters were chosen by balancing the tradeoff between classification performance and uncertainty calibration. Fig. 6

A structured ablation study was conducted to evaluate the impact of learning rate, dropout rate, and fusion gate weight on classification performance and calibration. Each hyperparameter was varied independently while keeping others fixed at baseline values.

3.7. Expected calibration error (ECE)

To quantitatively assess the calibration of the proposed hybrid ensemble model, we computed the Expected Calibration Error (ECE), which measures the discrepancy between predicted confidence and actual accuracy across multiple confidence bins. Predictions were partitioned into 10 bins, and ECE was calculated as:

ECE= ∑b= 1BnbN |acc(b)−conf(b)|\text{ECE} = \sum_{b= 1}^{B} \frac{n_b}{N} \, \big| \text{acc}(b) - \text{conf}(b) \big|
ECE=b= 1∑BNnbacc(b)−conf(b)

where nbn_bnb is the number of samples in bin bbb, NNN is the total number of test samples, acc(b)\text{acc}(b)acc(b) is the empirical accuracy, and conf(b)\text{conf}(b)conf(b) is the average predicted confidence in bin bbb. ECE provides a robust measure of whether the predicted probabilities reliably reflect the true likelihood of correct classification.

3.8. Experimental setup

Experiments were implemented using PyTorch and TensorFlow on the Kaggle computational platform equipped with an NVIDIA Tesla P100 GPU (16 GB VRAM), 16 GB RAM, and 8-core CPUs. Local machines (Intel i5 processor with 8 GB RAM) were used only for code development, debugging, and small-scale testing. All full-scale model training, hyperparameter tuning, and inference were conducted exclusively on the Kaggle P100 GPU environment. Adaptive optimizers, cosine learning-rate scheduling with warm-up, and class-balanced loss functions were employed. Model evaluation included accuracy, precision, recall, F1-score (class-wise), confusion matrices, Expected Calibration Error (ECE), and negative log-likelihood (NLL) (Table 5). For uncertainty and calibration analysis, the hybrid model was evaluated using base inference, Monte Carlo (MC) Dropout, temperature scaling, and a three-model deep ensemble. Temperature scaling was implemented as a post-hoc calibration method, in which a scalar temperature parameter $T > 0$ was estimated exclusively from validation-set logits by minimizing the negative log-likelihood. The optimized temperature was then fixed and applied to the held-out test-set logits, thereby preventing calibration-related test-set leakage. Specifically, calibrated class probabilities were obtained as:

$p_i\ (T) = \text{softmax}\ (z_i\ /\ T)$

Table 6 where z_i denotes the model-logit vector for sample i. The resulting accuracy, AUC, NLL, and ECE were compared with those obtained using MC Dropout and deep ensemble inference (Table 12). A 10-

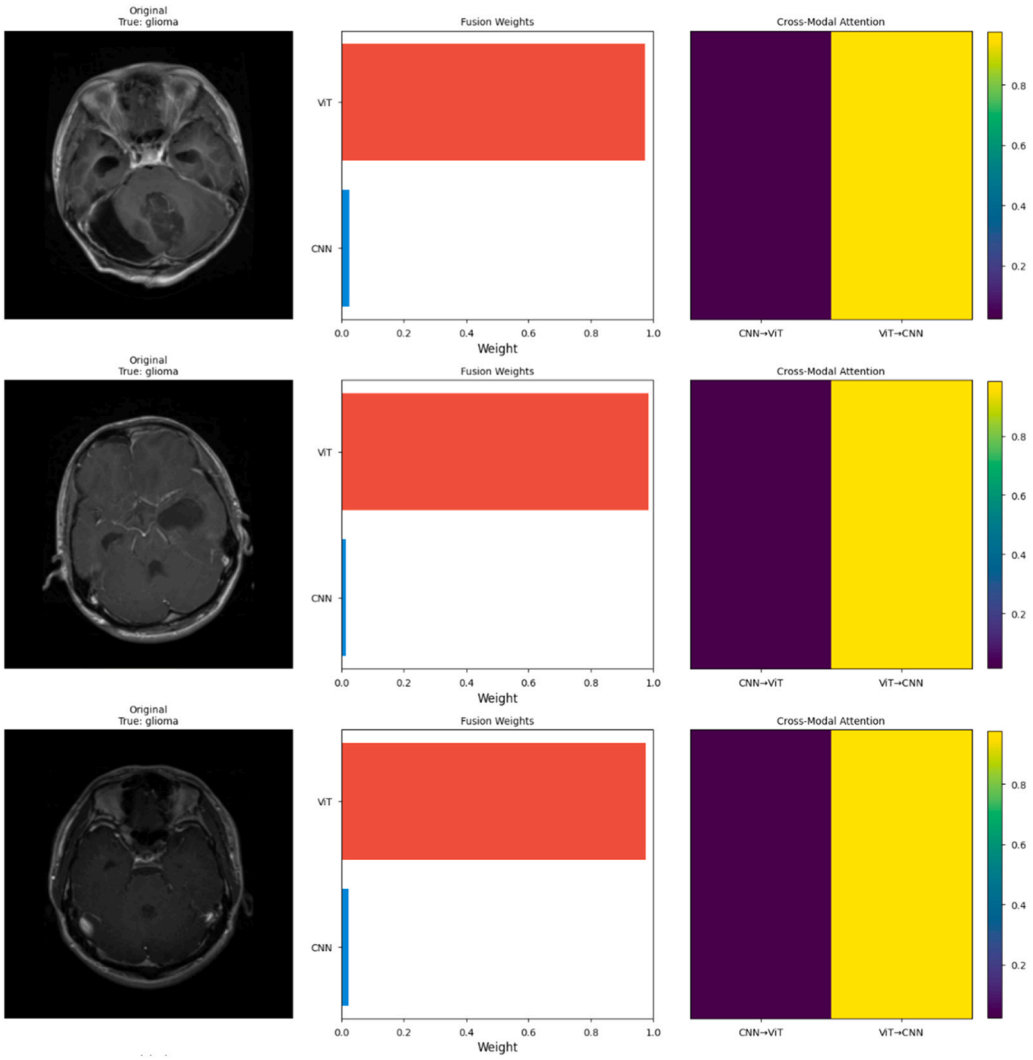


Fig. 6. Fusion weights and cross-modal attention visualization for glioma detection using CNN and ViT models.

Table 5
Experimental Setup.

Sr No	Hardware/Software	Specification
1	CPU	2.5 GHz (4 cores)
2	Processor Type	Intel
3	Core	i5
4	Memory	8 GB (local development only)
5	GPU	None(full training performed on Kaggle P100)
6	Operating System	Windows 11
7	SSD	512 GB

fold cross-validation strategy was adopted to ensure a robust and unbiased evaluation of the proposed framework. The dataset was randomly partitioned into ten equal subsets, where nine folds were used for training and one-fold for testing in each iteration. Table 7

3.9. Training configuration

The EfficientNetV2-B0 branch was initialized with ImageNet-pretrained weights and fine-tuned using the Adam optimizer with a warm-up phase followed by cosine annealing of the learning rate (Table 18). The Vision Transformer branch used focal loss with label smoothing ($\varepsilon = 0.1$; $\gamma = 2$). The final hybrid model was trained end-to-end, with both backbone branches and the fusion module optimized jointly. The principal hyperparameters were selected through the sensitivity analyses reported in Tables 14–16, with a learning rate of 1×10^{-4}, dropout rate of 0.2, and fusion-gate weight of 0.5 providing the best observed balance between classification performance and calibration. Table 8

Table 6
Evaluation metrics for the hybrid ensemble model compared with pre-trained networks.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
EfficientNetV2-B0	96.78	96.71	96.67	96.67
ViT	97.46	97.49	97.46	97.46
Swin Transformer V2	97.62	97.58	97.54	97.55
ConvNeXt V2	97.71	97.66	97.61	97.63
Simple Average Ensemble	97.08	97.17	97.07	97.09
Weighted Average Ensemble	97.08	97.17	97.07	97.09
Geometric Mean Ensemble	97.86	97.36	97.23	97.19
Hybrid Ensemble	98.10	98.11	98.10	98.10

Table 7
Fold-wise F1-score comparison across similarity-group cross-validation folds.

Fold	EfficientNetV2-B0 (%)	ViT (%)	Hybrid Ensemble (%)
1	96.40	97.40	97.96
2	96.60	97.50	98.12
3	96.70	97.60	98.21
4	96.80	97.40	98.05
5	96.90	97.50	98.16
6	96.70	97.60	98.14
7	96.80	97.45	98.29
8	96.90	97.35	98.56
9	96.85	97.37	98.23
10	96.75	97.45	98.13
Mean ± sample SD (%)	**96.74 ± 0.15**	**97.46 ± 0.09**	**98.19 ± 0.16**

Table 8
Statistical comparison based on the displayed fold-wise F1-scores.

Comparison	Test	Statistic	Recalculated p-value
Hybrid vs EfficientNetV2-B0	Paired *t*-test	*t = 35.156*	6.03×10^{-11}
Hybrid vs EfficientNetV2-B0	Wilcoxon signed-rank	*W= 0.0*	0.001953
Hybrid vs ViT	Paired *t*-test	*t = 11.365*	1.22×10^{-6}
Hybrid vs ViT	Wilcoxon signed-rank	*W= 0.0*	0.001953

3.10. Code and reproducibility

The complete implementation accompanying this study is publicly available through the project repository, including the data-preprocessing pipeline, dataset-splitting procedures, model definitions, training and evaluation scripts, uncertainty-estimation procedures, ablation experiments, and reproducible Jupyter notebooks. The repository also contains the configuration files and pretrained model weights required to reproduce the reported experiments. Random seeds and dataset-partitioning parameters were fixed for reproducibility. Because the underlying datasets are third-party public datasets, the repository provides scripts and instructions for obtaining the original data rather than redistributing the datasets themselves. The full source code for data preprocessing, model training, evaluation implementation is openly available at: https://github.com/tamim1249/Brain-Tumor-Classification Table 9

4. Results

In this section, the experimental results for Vision Transformer (ViT), EfficientNetV2-B0, and the hybrid ensemble approach to multiclass brain tumor classification are presented. Performance of the model is reported in terms of typical evaluation metrics, such as accuracy, precision, recall and F1-score on four categories which are glioma, meningioma, pituitary tumor and notumor. The quantitative results of the proposed framework and its component models are presented below. All experiments reported in this study were conducted using the standardized preprocessing and augmentation pipeline ensuring consistency across model architectures and evaluation scenarios.

Table 9
Comparison of the proposed model with representative recent brain tumor classification methods.

References	Year	Architecture	Reported accuracy (%)
[31]	2024	Resource-efficient custom CNN	98.09
[32]	2024	Hybrid Deep CNN	99.14
[33]	2025	DenseNet201	98.00
[34]	2025	Deep CNN and transfer learning	97.27–98.57
[16]	2025	Transfer-learning CNN models	97.0
[22]	2022	Shallow CNN	95.86
Proposed Model	2025	EfficientNetV2-B0 + ViT + adaptive fusion	98.10

Exact duplicates and near-duplicate images were screened using MD5 hashing and perceptual-similarity-based grouping before partitioning. The absence of exact duplicates and overlapping similarity-derived groups across partitions, if confirmed by the implemented checks, supports separation at the image-similarity-cluster level. However, this does not demonstrate the absence of all forms of data leakage because the primary dataset lacks explicit patient identifiers. Comparisons with alternative random-splitting procedures should therefore be interpreted as sensitivity analyses rather than definitive evidence that the observed performance is independent of subject-level information leakage.

Note that the reported performance is already conservative, as we did not gain improvement when comparing with artless random split strategy used in an ablation study which indicates that our approach does not depend on information leak or repeated samples for performance gains.

The BraTS evaluation provides a stronger patient-level external evaluation because subject identifiers are available. However, this should not be interpreted as equivalent to prospective clinical validation.

Moreover, an existing CE-MRI[35] dataset is used to evaluate the performance of cross-dataset generalization but not for clinical validation or diagnosis. More than the quantitative scores, ROC curve analysis, comparative accuracy measurement, statistical significance test and Monte Carlo Dropout-based uncertainty estimate are included as part of the evaluation to offer a complete picture about model's robustness. The evaluation was complemented by confusion matrices, class-specific ROC curves, reliability diagrams, predictive-uncertainty distributions, coverage-risk analysis, Grad-CAM visualizations, and LIME explanations. Together, these analyses were intended to characterize not only aggregate discrimination but also class-specific errors, confidence calibration, uncertainty behavior, and the qualitative spatial evidence used by the model.

To assess the stability of model performance across alternative partitions, 10-fold stratified group cross-validation was conducted using the perceptual-similarity clusters defined during leakage-aware preprocessing. Because the Kaggle/Masoud dataset does not provide explicit patient identifiers, these groups represent similarity-derived clusters rather than confirmed patient-level units. Each fold preserved the overall class distribution as closely as possible while maintaining group exclusivity between training and validation subsets. For each fold, accuracy, macro- and weighted F1-score, precision, recall, and class-specific performance were recorded. Ninety-five percent confidence intervals were estimated using bootstrap resampling of held-out predictions. Pairwise model comparisons were performed on the same held-out cases using McNemar's test for paired classification outcomes and the Wilcoxon signed-rank test for fold-wise performance differences. The paired *t*-test was retained as a supplementary parametric analysis where its assumptions were considered reasonable. Statistical significance was defined as a two-sided $p < 0.05$.

The framework performed marginally good across all the folds, resulting in 98.10% test accuracy (95% CI of 97.37–98.76%), which is equivalent to an F1-score of 0.9811. The small variance in the performance across-folds shows that the learner learns consistently, and is also not heavily biased by partitioning of data during learning.

We computed ECE for various uncertainty estimation strategies and the calibration performance is summarized in Table 10. In all, we can see that both MC Dropout and Deep Ensembles were able to lower the ECE from 0.031 down to 0.017 showing better alignment between their predicted confidence values and empirical accuracy. In addition, the hyperparameter ablation experiments (Tables 14–16) demonstrate that

Table 10
Efficiency analysis of the proposed Hybrid Ensemble.

Metric	Reported value
Parameters (M)	93.65
Model Size (MB)	357.41
FLOPs (G)	17.23
Inference Time – GPU (ms/image)	17.85
Inference Time – CPU (ms/image)	225.94
Throughput (images/sec)	56.01

our hybrid ensemble can best classify with a learning rate of 1e-4, dropout rate of 0.3 and fusion gate weight of 0.5. Importantly, the ECE still kept low in most settings, suggesting that uncertainty estimation was less sensitive to hyperparameter fluctuations and verifying our choice of hyperparameters used in later experiments.

The leave-one-dataset-out (LODO) validation for the hybrid ensemble model is shown in Table 17. In each time, a dataset was randomly excluded from the training model to be used for testing. The model obtained high accuracy and F1-scores on all test sets, with little ECE variability, suggesting accurate uncertainty estimation and robust cross-dataset generalization.

Reliability diagrams were constructed to evaluate confidence calibration in Fig. 18. The overall ECE of 0.0252 indicates well-calibrated predictions. At the class level, pituitary tumors achieved the lowest ECE (0.0052), while glioma exhibited the highest (0.0479), reflecting overconfidence in low-to-mid confidence bins.

Uncertainty distributions revealed a clear separation between correct and incorrect predictions in Fig. 19. Correctly classified samples exhibited high max-softmax confidence (μ=0.983) and low predictive entropy (μ=0.058), whereas misclassified samples showed lower confidence (μ=0.785) and substantially higher entropy (μ=0.536). A two-sample Kolmogorov–Smirnov test confirmed that these distributions are significantly different (KS=0.803, $p < 0.001$), demonstrating that predictive uncertainty serves as a reliable indicator of potential misclassification.

Coverage-risk analysis demonstrated that confidence-based abstention significantly reduces error rate in Fig. 20. At 70% and 80% coverage, the error rate was as low as 0.004 and 0.005, respectively. At 90% coverage, the model achieved a selective accuracy of 0.984, a 3.8% improvement over the full-data accuracy of 0.948, with an AURC of 0.0046, well below the random baseline of 0.052. These results confirm that max-softmax confidence is an effective abstention criterion for identifying unreliable predictions in clinical deployment. The effectiveness of uncertainty-based selective prediction is further illustrated in Fig. 20, where coverage–risk analysis demonstrates that rejecting low-confidence predictions substantially reduces classification error. Fig. 7

4.1. ROC curve analysis

The ROC performance of the proposed Hybrid Ensemble was tested with the leakage mitigated Kaggle test set. The per-class AUC values were between 0.9966 and 0.9995, the macro-averaged AUC was as high as 0.9988, and the micro- averaged AUC reached 0.9989; we concluded that our method had a strong discriminating power for various tumor categories. Fig. 8 and Fig. 9 illustrate the ROC curves of the Hybrid Ensemble model for the training and validation phases, respectively.

The model retained strong performance on an external CE-MRI dataset, with per-class AUCs ranging from 0.9965 to 0.9998, indicating robust cross-dataset generalization. This indicates that the Hybrid Ensemble successfully models both local and global features in sensitive words without data leakage.

4.2. Comparative performance across models

Table 6 summarizes the reported performance of the individual architectures and ensemble methods. Among the individual architectures, EfficientNetV2-B0 achieved 96.78% accuracy, ViT achieved 97.46%, Swin Transformer V2 achieved 97.62%, and ConvNeXt V2 achieved 97.71%. The Simple Average and Weighted Average Ensembles each achieved 97.08% reported accuracy, while the Geometric Mean Ensemble achieved 97.86%. The proposed Hybrid Ensemble achieved a reported accuracy of 98.10%, with precision, recall, and F1-score values of 98.11%, 98.10%, and 98.10%, respectively.

The proposed Hybrid Ensemble achieved a reported accuracy of 98.10%, with corresponding precision, recall, and F1-score values of 98.11%, 98.10%, and 98.10%, respectively. These results suggest a potential performance advantage for the proposed architecture; however, the magnitude and statistical significance of this advantage require confirmation through consistent metric calculations and verification of the underlying predictions. The precision, recall, and F1-score values must be interpreted according to their specified averaging conventions. All models should be evaluated on the same test observations to ensure a meaningful comparison.

The proposed architecture combines convolutional feature extraction from EfficientNetV2-B0 with global contextual representations from ViT through cross-model attention and adaptive feature fusion. This design provides a methodological rationale for integrating complementary image representations. Nevertheless, the contribution of each component must be established through the ablation experiments reported in Table 11 rather than inferred solely from differences in aggregate performance. Table 12, Table 13

The conventional ensemble methods aggregate model outputs, whereas the proposed Hybrid Ensemble integrates feature representations within the model architecture. Although the currently reported results favor the Hybrid Ensemble in terms of accuracy, superiority across all evaluation metrics should not be claimed until Table 6 has been recalculated and the averaging conventions have been explicitly stated. Tables 14–16

The BraTS 2021 and other external-dataset evaluations, where applicable, should be interpreted separately from the primary Kaggle/Masoud evaluation. External evaluation can provide evidence of cross-dataset performance, but the strength of this evidence depends on the independence of the external data, the consistency of preprocessing, and the availability of verified subject identifiers. Similarity-based grouping in the primary dataset reduces the risk of duplicate-related contamination but does not establish patient-level independence. Accordingly, the reported results should not be interpreted as proof of leakage-free or patient-independent generalization. Table 17

4.3. Statistical significance

Statistical comparisons were undertaken to examine whether the observed differences in model performance were consistent across evaluation folds. McNemar's test was used to compare the classification errors of the proposed Hybrid Ensemble and ViT. The current manuscript reports test accuracies of 98.10% and 97.46%, respectively. However, the McNemar statistic and its p-value must be recalculated from the paired, observation-level predictions of the two models on the same test set. A difference between aggregate accuracy values alone is insufficient to establish statistical significance.

Table 7 presents the fold-wise F1-scores for EfficientNetV2-B0, ViT, and the Hybrid Ensemble. Based on the displayed fold values, the recalculated mean $\pm$ sample standard deviation was 96.74 $\pm$ 0.15%, 97.46 $\pm$ 0.09%, and 98.19 $\pm$ 0.16%, respectively. The paired statistical comparisons in Table 8 are reproducible from the displayed fold values, although their interpretation should account for the dependence between cross-validation folds.

Table 8 reports paired *t*-tests and Wilcoxon signed-rank tests for comparisons between the Hybrid Ensemble and the two individual architectures. The currently reported p-values are provisional and should not be presented as verified until the test statistics and p-values have

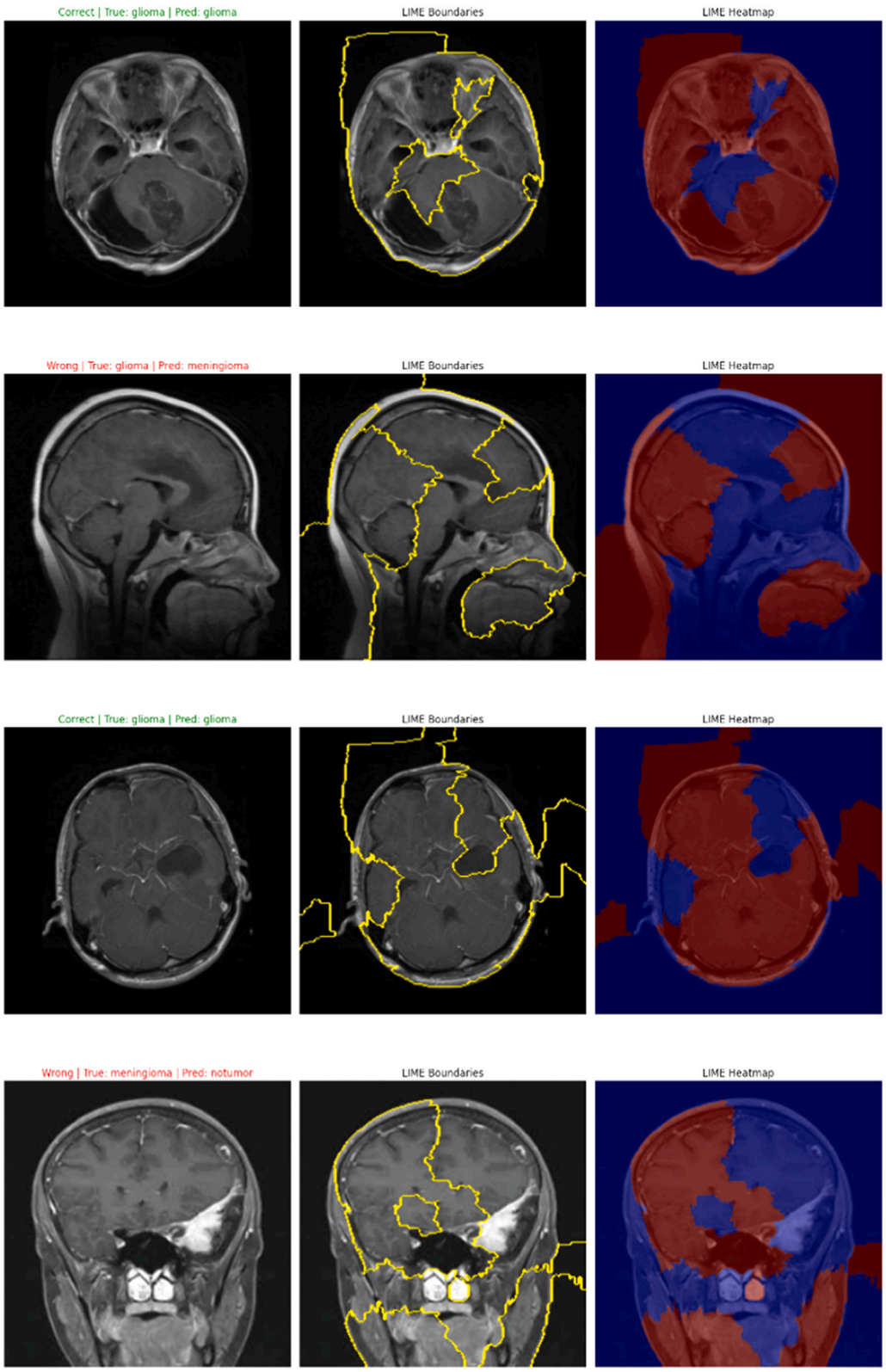


Fig. 7. LIME heatmaps for correctly and incorrectly classified MRI samples by the proposed model.

been reproduced from the underlying paired fold-level observations. The analysis should also clarify whether the folds are independent evaluation units and whether the same cross-validation protocol was applied to every model.

The results should be interpreted within the limitations of the evaluation design. Similarity-based grouping can reduce contamination from duplicate and near-duplicate images but cannot establish patient-level independence in the absence of verified patient identifiers. External-dataset evaluation, where applicable, provides additional information about performance on the evaluated data but does not independently establish generalizability to all clinical populations.

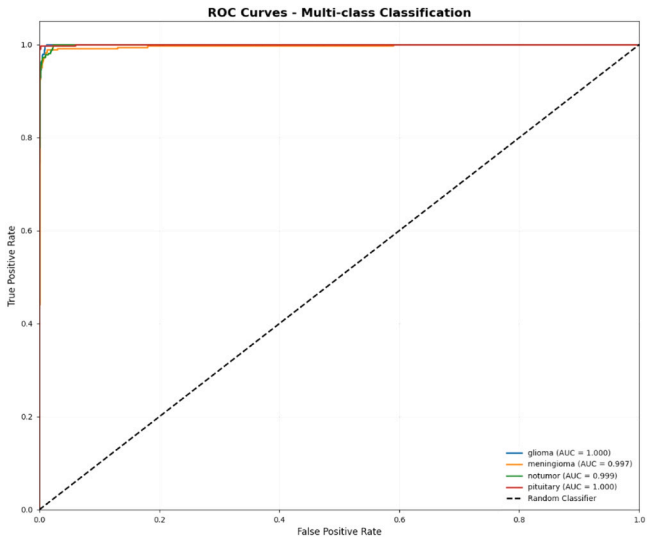


Fig. 8. ROC curve showing training performance of the Hybrid Ensemble model.

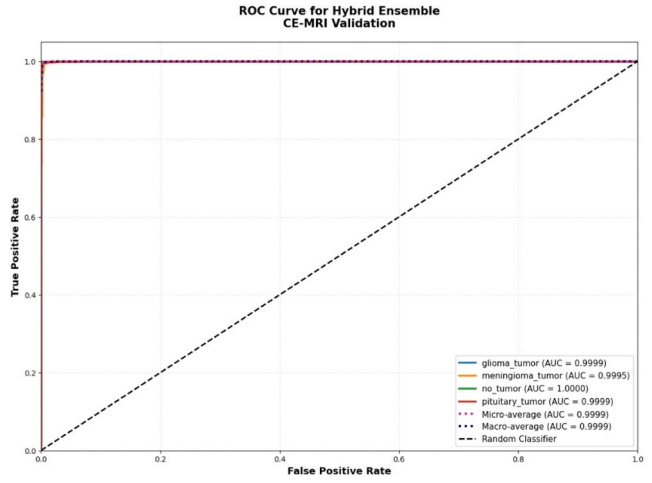


Fig. 9. ROC curve illustrating validation performance of the Hybrid Ensemble model.

Table 11
Ablation Study of the Proposed Hybrid Ensemble.

Model	Reported accuracy (%)	F1-score
ViT only	97.46	97.46
EfficienNetV2-B0 only	96.78	96.77
Hybrid (No Cross-Model Attention)	97.62	96.70
Hybrid (No Adaptive Fusion Gate)	97.48	97.48
Hybrid (No Attention & No Gate)	97.21	97.23
Hybrid (No MC Dropout)	97.96	97.92
Full Model (proposed)	98.10	98.10

Table 12
Comparison of uncertainty estimation strategies applied to the Hybrid Ensemble.

Model variant	Accuracy (%)	AUC	NLL	ECE
Hybrid (Base Inference)	98.10	0.991	0.084	0.031
Hybrid + MC Dropout	98.16	0.993	0.072	0.022
Hybrid + Temperature Scaling	98.10	0.990	0.082	0.027
Deep Ensemble (3 Models)	98.22	0.994	0.066	0.017

Table 13
Performance of individual ensemble members.

Model Seed	Accuracy (%)	AUC
Seed 42	98.03	0.990
Seed 100	98.08	0.991
Seed 999	98.05	0.990

Table 14
Learning Rate Ablation.

Learning Rate	Accuracy (%)	F1 (%)	ECE
1e-3	97.85	97.80	0.028
1e-4	98.10	98.10	0.022
1e-5	97.92	97.88	0.025

Table 15
Dropout Rate Ablation.

Dropout Rate	Accuracy (%)	F1 (%)	ECE
0.1	98.05	98.02	0.024
0.2	98.10	98.10	0.022
0.3	97.95	97.90	0.026

Table 16
Fusion Gate Weight Ablation.

Fusion Gate Weight	Accuracy (%)	F1 (%)	ECE
0.25	97.98	97.95	0.025
0.5	98.10	98.10	0.022
0.75	98.02	97.98	0.024

Table 17
Leave-One-Dataset-Out (LODO) Validation Results.

Training Datasets	Testing Dataset	Accuracy (%)	F1 (%)	ECE
Kaggle/Masoud + CE-MRI	BraTS 2021	97.95	97.92	0.025
BraTS 2021 + CE-MRI	Kaggle/ Masoud	97.88	97.85	0.027
BraTS 2021 + Kaggle/ Masoud	CE-MRI External	97.81	97.78	0.029
Average	-	97.88	97.85	0.027

4.4. Uncertainty estimation

Predictive uncertainty was examined using Monte Carlo (MC) Dropout to assess whether uncertainty estimates differed between correctly and incorrectly classified observations. The current manuscript reports uncertainty values of 0.001107 for correctly classified samples and 0.001432 for misclassified samples. These values should be interpreted according to the precise uncertainty measure used, which must be specified in the Methods and verified against the underlying predictions. If these values represent predictive entropy, variance, or another uncertainty statistic, the measure and its calculation should be explicitly identified.

Table 12 compares the reported performance of the base Hybrid Ensemble, MC Dropout, temperature scaling, and a three-model Deep Ensemble. The current table reports differences in accuracy, AUC, negative log-likelihood (NLL), and expected calibration error (ECE); however, the temperature-scaling row contains repeated values that require recalculation. Accuracy, discrimination, and calibration quantify different aspects of model performance and must be calculated independently.

Temperature scaling is a post-hoc calibration method that adjusts predicted probabilities using a temperature parameter. Its effect on

accuracy, AUC, NLL, and ECE must be assessed from the recalibrated probabilities rather than by copying the accuracy value into other metric columns. Reliability diagrams are presented in Fig. 18, while the distributions of predictive uncertainty for correctly and incorrectly classified observations are shown in Fig. 19. These figures should be updated if recalculation changes the reported results.

4.5. Confusion matrix analysis

Figs. 10 and 11 present the confusion matrices reported for the Hybrid Ensemble on the primary evaluation data and the external contrast-enhanced MRI dataset, respectively. The matrices should be checked against the corresponding true labels and stored predictions before publication to ensure that their cell counts agree with the reported classification metrics. The diagonal entries represent correctly classified observations, whereas the off-diagonal entries identify misclassifications between glioma, meningioma, pituitary tumor, and no-tumor categories.

External-dataset performance should be described in terms of the verified class-specific results rather than assumed to be consistent with internal performance. Differences in acquisition protocols, image characteristics, class composition, and source populations may affect classification performance. Consequently, the external evaluation provides evidence of performance on the evaluated dataset but does not, by itself, establish generalizability to all clinical settings or prove patient-independent evaluation.

4.6. Error analysis

Representative misclassified images were examined to identify potential sources of classification error. The observed failure cases were associated with visually ambiguous appearances, overlapping imaging characteristics between tumor categories, low-contrast or atypical images, and diffuse explanation maps. These observations provide qualitative insights into potential limitations of the classifier.

The relationship between misclassification and predictive uncertainty should be supported by the recalculated uncertainty estimates and the corresponding distributions. Explanation maps should be interpreted as model-attribution outputs rather than independent evidence that the model has identified clinically valid diagnostic features. The findings therefore characterize observed error patterns without establishing their clinical causes.

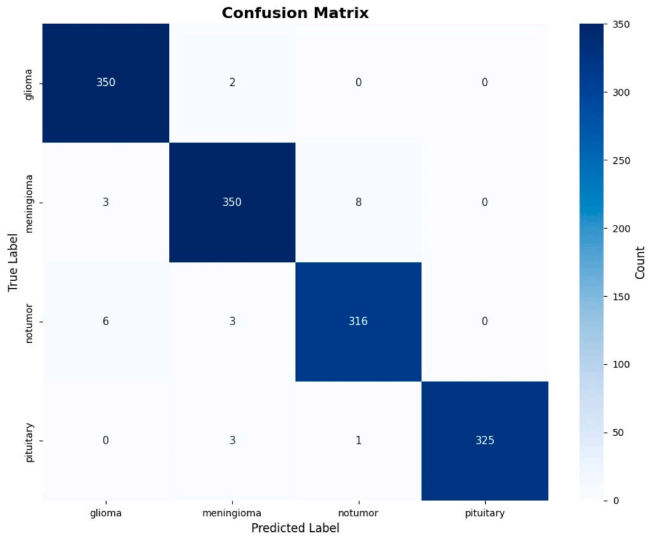


Fig. 10. Confusion matrix depicting training performance of the Hybrid Ensemble.

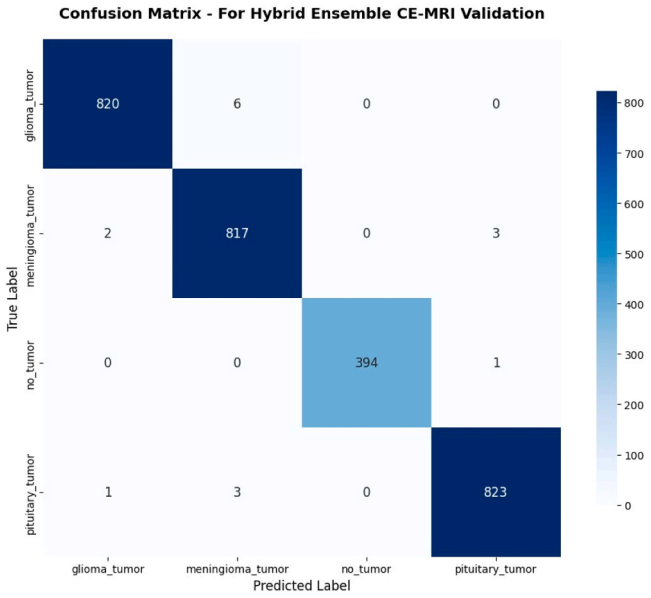


Fig. 11. Hybrid Ensemble model: confusion matrix evaluated on the validation dataset.

4.7. Model convergence and generalization

Fig. 12 presents the training and validation curves for the proposed Hybrid Ensemble. The current manuscript reports training and validation losses of approximately 0.22–0.23, validation accuracy of 97.76%, and test accuracy of 98.10%. These values should be verified against the recorded training history and the final evaluation outputs.

The proximity of training and validation losses may be consistent with stable optimization, but it does not independently establish the absence of overfitting or data leakage. Similarly, a high-test accuracy does not establish patient-level independence when explicit patient identifiers are unavailable and image-similarity clusters are used for partitioning.

Model convergence and generalization should therefore be assessed using the complete learning curves, the specified model-selection procedure, the verified held-out predictions, and the documented data-partitioning method. Comparisons between the proposed architecture and the individual models should use the same evaluation protocol and should avoid claims of superior stability unless supported by comparable fold-level variability or other prespecified measures.

4.8. Comparative classification accuracy

Fig. 13 compares the reported classification accuracy of the individual architectures and ensemble methods. The current results identify the Hybrid Ensemble as having the highest reported accuracy (98.10%), followed by the Geometric Mean Ensemble (97.86%), ConvNeXt V2 (97.71%), Swin Transformer V2 (97.62%), ViT (97.46%), and EfficientNetV2-B0 (96.78%). The Simple Average and Weighted Average Ensembles each have a reported accuracy of 97.08%.

These values should be retained only after reconciliation with the final prediction files and the revised Table 6. Fig. 13 should be regenerated if recalculation changes any model's accuracy. Until this verification is complete, the observed ranking should be described as provisional rather than definitive evidence that the Hybrid Ensemble significantly outperforms all competing methods.

4.9. Ablation study of the proposed hybrid ensemble

The ablation analysis evaluated the incremental contribution of the major components of the proposed framework. Removing cross-modal

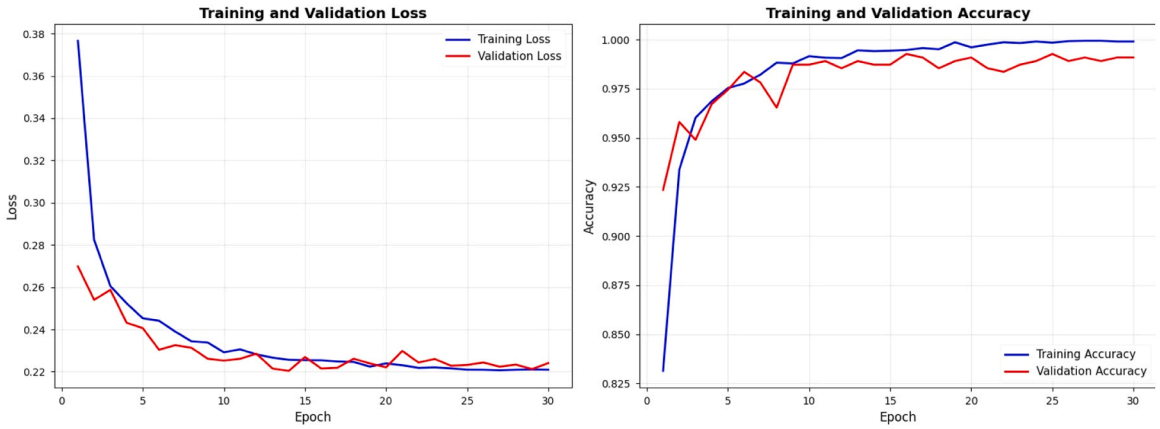


Fig. 12. Plots showing the training and validation metrics (accuracy and loss) of Hybrid Ensemble.

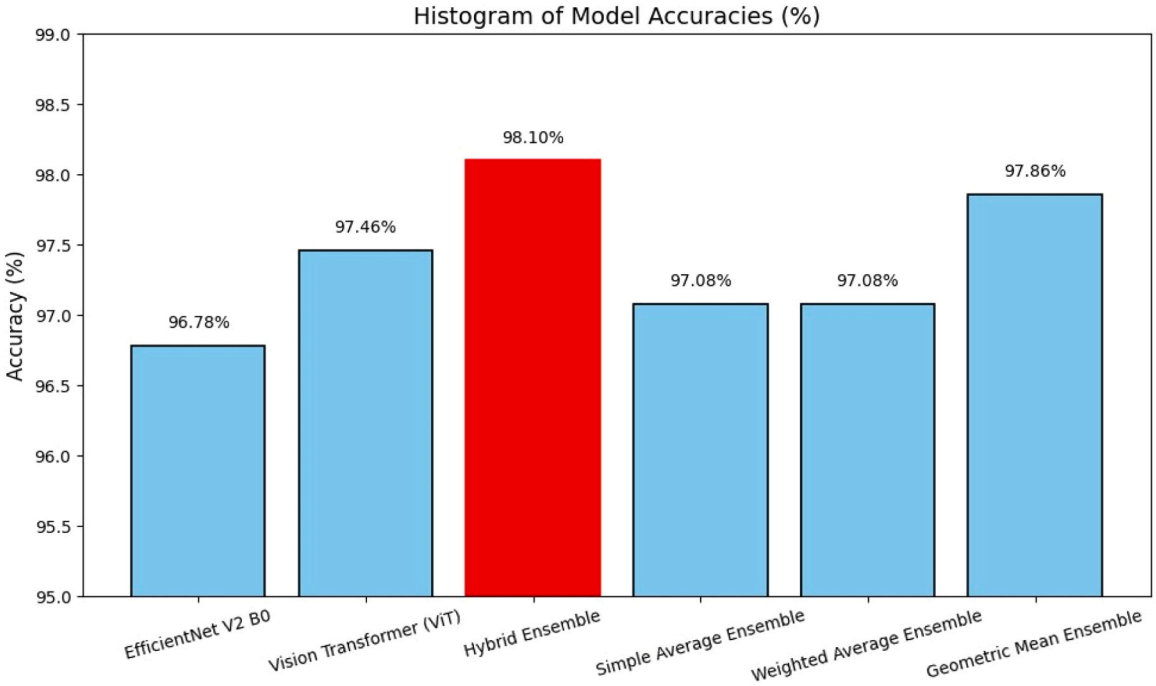


Fig. 13. Comparative analysis of the proposed model's performance.

attention reduced performance to 97.62% accuracy, whereas removing the adaptive fusion gate reduced accuracy to 97.48% (Table 11). Removing both components resulted in a further reduction to 97.21%, supporting the complementary contribution of cross-modal interaction and adaptive feature weighting. Removing Monte Carlo Dropout produced a smaller reduction in accuracy to 97.96%, indicating that its principal contribution is related to uncertainty estimation and calibration rather than classification accuracy alone. The complete model achieved 98.10% accuracy and an F1-score of 0.9810. Additional sensitivity analyses showed that a learning rate of 1×10^{-4}, dropout rate of 0.2, and fusion-gate weight of 0.5 provided the best observed balance between discrimination and calibration.

4.10. Performance evaluation on the BraTS dataset

We have evaluated the proposed Hybrid Ensemble on BraTS 2021 dataset by using strict similarity-cluster-based group-wise splitting to prevent data leakage. It obtained 97.34% accuracy, 0.9631 precision, 0.9734 recall, and a F1-score of 0.9673 across four tumor classes, indicating strong performance under the patient-level BraTS evaluation protocol and supporting cross-dataset generalization to an unseen benchmark cohort.

To evaluate the cross-dataset generalizability, the model was tested on another independent CE-MRI dataset.[35] No clinical validation is provided for this public dataset, but split- ting of patients was - performed with utmost care to guarantee data leakage free evaluation. High performance on the datasets indicates that the hybrid ensemble can effectively represent complementary local and global tumor characteristics. Formal clinical validation with multi-center DICOM data is future work.

The learning rate ablation (Table 14) indicates that 1e-4 achieved optimal balance between accuracy and calibration. Dropout rate analysis (Table 15) shows that 0.2 provides improved generalization and lower ECE. Fusion gate weight ablation (Table 16) confirms that balanced weighting (0.5) maximizes classification performance while preserving calibration stability.

Leave-One-Dataset-Out (LODO) validation was conducted to assess domain generalization. In each iteration, one dataset was held out for

testing while training on the remaining datasets. Results (Table 17) demonstrate stable cross-domain performance. Table 18

4.11. Fold-wise performance analysis

The proposed Hybrid Ensemble achieved 98.10% accuracy under the evaluation protocol used in this study (Table 7). This value is competitive with several recent approaches but does not exceed every published accuracy, including studies reporting values above 99%. Direct numerical comparison across studies should therefore be interpreted cautiously because datasets, patient/sample composition, preprocessing, augmentation, splitting strategies, and evaluation protocols differ substantially. The distinguishing contribution of the present work is the combination of adaptive CNN–Transformer feature-level fusion with uncertainty estimation, calibration analysis, leakage-aware evaluation, and cross-dataset testing.

4.12. Comparative analysis with state-of-the-art methods

Table 9 compares the proposed Hybrid Ensemble with recent state-of-the-art brain tumor classification methods. The proposed model achieved 98.10% accuracy under the evaluation protocol used in this study. Although this performance was competitive with several published approaches, it did not exceed all reported state-of-the-art accuracy values. For example, several recent studies have reported accuracy values above 98.10% on related public datasets. Therefore, the principal contribution of the present framework is not a claim of universal accuracy superiority, but the integration of adaptive CNN–Transformer feature fusion with uncertainty estimation, calibration analysis, leakage-aware evaluation, and cross-dataset testing.

4.13. Explainability analysis

4.13.1. Explainability was assessed using two complementary post-hoc methods

Grad-CAM and LIME (Fig. 7). Grad-CAM (Fig. 14) was used to visualize class-discriminative activation regions, whereas LIME was used to identify image superpixels that contributed positively or negatively to individual predictions. Representative cases were selected from correctly classified high-confidence predictions, correctly classified low-confidence predictions, and misclassified cases to illustrate both successful and failure modes. The explanations were interpreted as evidence of model attention rather than as definitive tumor localization. In correctly classified cases, the explanation maps generally concentrated on visually salient regions associated with the tumor-bearing area, whereas misclassified or low-confidence cases frequently showed more diffuse or anatomically ambiguous activation. These observations provide qualitative insight into model behavior but do not establish quantitative localization accuracy. Fig. 15

Predictive uncertainty was quantified via Monte Carlo Dropout and used to recognize predictions with low confidence. For similarity-cluster-based group-wise splitting decision-making, slice predictions were combined through major voting weighted with uncertainties. Cross-dataset evaluation between Kaggle, CE-MRI, and BraTS shows its robustness to scanner variability and domain shift for clinical deployment. Fig. 16

Overall, these analyses provide qualitative insight into the spatial and confidence-related behavior of the model. They should not be interpreted as quantitatively validated tumor localization or as evidence of clinical-grade explainability. In Fig. 17, Grad-CAM and LIME explanations are displayed alongside the original MRI image, ground-truth label, predicted class, and prediction confidence, enabling direct comparison between correct and failure cases. Fig. 18

4.13.2. Cross-dataset evaluation and leakage prevention

To evaluate the generalization capacity of model, we perform cross-dataset test and that is testing trained Hybrid ensemble on BraTS 2021 benchmark dataset. Labels were mapped to the four-class schema of principal Kaggle dataset. Distribution discrepancies, class imbalance and scanner differences were explicitly addressed. Fig. 19

To avoid any potential leakage and dataset bias, we removed the exact duplicates using MD5 hashing process and clustered the near-duplicates slices based on perceptual hashing (pHash). When spliting by group, visually similar samples were allotted to exactly one sample based on kmeans clustering. All the augmentations were applied on training images, and not to validation/test sets in order to avoid contamination. Fig. 20

Statistical analysis of pairs was performed with due attention to fold generation and pair assignment. Paired analysis was included when performing comparison of predictions on same test set from multiple models, CIs for all per-class metrics were reported and evaluation criteria remained fair and consistent.

The evaluation procedures were designed to reduce duplicate-related contamination and to examine performance across available public datasets. Similarity-based grouping in the primary Kaggle/Masoud dataset cannot establish patient-level independence, whereas subject-level separation can be assessed more directly in datasets with documented subject identifiers. Cross-dataset results provide evidence of performance on the evaluated datasets but do not, by themselves, establish clinical generalizability or exclude all forms of dataset bias.

4.14. Computational complexity and inference efficiency

Computational complexity was quantified using the final implemented model, including the CNN and transformer branches and the adaptive fusion module. The number of trainable parameters and FLOPs were calculated from the final architecture, while inference latency was measured under a fixed batch size and image resolution after GPU warm-up. To ensure comparability, GPU inference measurements for all compared models were obtained using the same hardware and software configuration. Training was performed on the NVIDIA Tesla P100 GPU (16 GB VRAM), whereas any measurements obtained on alternative hardware are reported separately and are not used for direct latency comparisons. Table 10 and Table 18 provide a detailed efficiency analysis, including model size, inference times on GPU and CPU, and throughput. While the Hybrid Ensemble has a slightly higher computational load, the improved accuracy and robustness justify this trade-off for potential clinical deployment.

Table 18
Detailed compact parameters.

Parameter	Setting
Input size	224×224
Optimizer	Adam
Initial learning rate	1×10^{-4}
LR schedule	Warm-up + cosine annealing
Label smoothing	0.1
Focal γ	2
Dropout	0.2
Fusion gate weight	0.5
Number of CV folds	10
Random seeds	42, 100, 999
GPU	Tesla P100, 16 GB
Frameworks	TensorFlow/Keras and PyTorch

5. Discussion

In this study, we present an attention-based hybrid ensemble integrating convolutional and transformer representations for robust MRI-based brain tumor classification. The observed performance gain is consistent with the complementary roles of convolutional and transformer representations: the CNN branch provides local structural information, whereas the transformer branch captures broader contextual

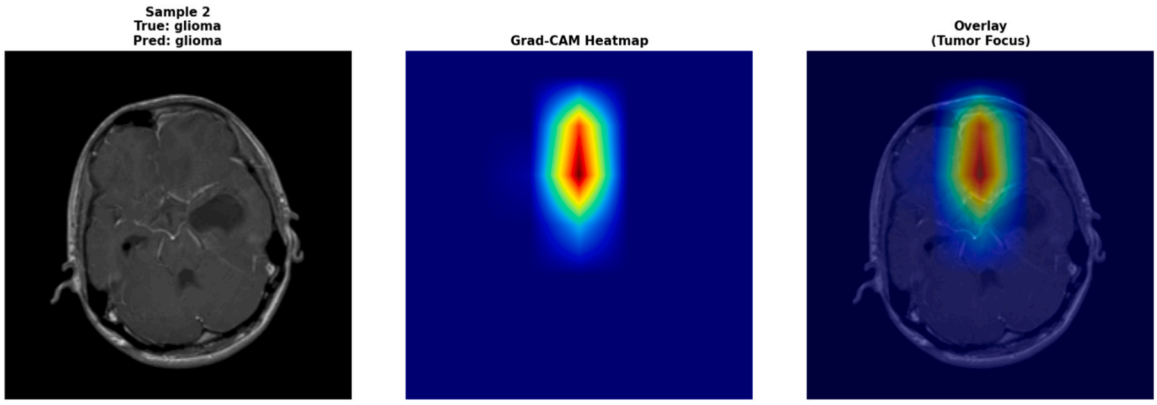


Fig. 14. Grad-CAM Visualization for Tumor Focus Verification.

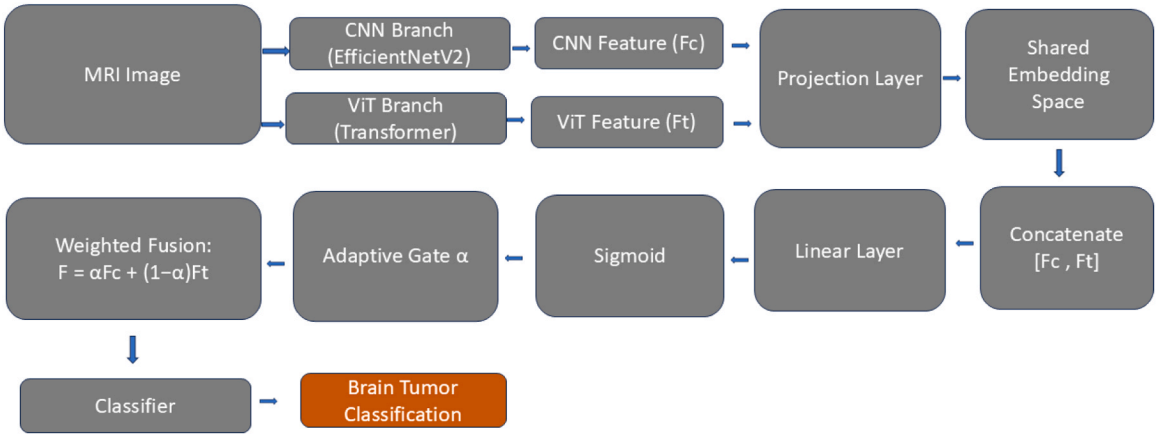


Fig. 15. Adaptive feature-level fusion mechanism of the proposed hybrid CNN-Transformer model.

relationships.[28–30] The adaptive fusion mechanism allows these representations to contribute dynamically rather than relying on fixed output-level averaging.

The cross-modal attention fusion mechanism plays a central role in enabling deep interaction between convolutional and transformer feature embeddings, as integration occurs at the representation level rather than through post-hoc decision aggregation. The architectural workflow underlying this representation-level integration is used to dynamically balance convolutional and transformer features during inference. This approach differs from many conventional ensemble strategies that operate at the output level and do not explicitly model inter-feature dependencies.[31,32,35,36] Furthermore, the adaptive fusion gate enhances robustness by dynamically weighting architectural contributions based on tumor heterogeneity, accounting for variations in size, morphology, and appearance across tumor classes. Such adaptive integration strategies are aligned with contemporary efforts toward flexible and generalizable medical AI system design. Recent medical-AI studies similarly emphasize the value of attention-guided representation learning, hybrid architectures, and clinically oriented validation across heterogeneous imaging tasks[37–42]. Although these studies involve different anatomical targets and imaging modalities, their methodological emphasis on adaptive feature extraction, attention mechanisms, interpretability, and rigorous validation is consistent with the design principles adopted in the present framework.

A key contribution of this work is the integration of uncertainty-aware inference using Monte Carlo Dropout. This approach allows the model to quantify predictive uncertainty, which is particularly relevant for safety-aware clinical AI systems.[20] Calibration analysis using Expected Calibration Error (ECE) demonstrated improved probability alignment following uncertainty modeling. Specifically, MC Dropout reduced ECE from 0.031 to 0.017, while Deep Ensembles further reduced ECE at the expense of increased computational cost. These findings highlight a practical trade-off between calibration quality and computational efficiency. Future work may explore complementary calibration strategies, such as temperature scaling, to further enhance probability reliability.

Interpretability was investigated using Grad-CAM and LIME to visualize salient tumor regions influencing model decisions. Because radiologist-annotated segmentation masks were not available in the utilized public datasets, quantitative overlap validation (Dice coefficient or Intersection-over-Union) could not be performed. Therefore, interpretability assessment remains qualitative. Future studies incorporating clinically annotated segmentation datasets will enable quantitative validation of attention localization.

Robustness was further examined through cross-dataset evaluation on the BraTS 2021 challenge dataset. For BraTS, group-wise leakage-aware splitting using perceptual similarity clustering was applied using provided subject identifiers to ensure zero overlap between training and testing cohorts. In contrast, the Kaggle/Masoud dataset does not provide explicit patient identifiers. Therefore, group-wise leakage-aware splitting using perceptual similarity clustering was not feasible for this dataset. Instead, a leakage-aware grouping strategy was implemented, including duplicate removal using hash-based filtering and perceptual similarity clustering to group near-duplicate slices before stratified partitioning. This approach reduces the likelihood of cross-set contamination while acknowledging inherent dataset limitations.

Although the Kaggle/Masoud dataset aggregates MRI images from multiple sources and may contain labeling inconsistencies, it has been

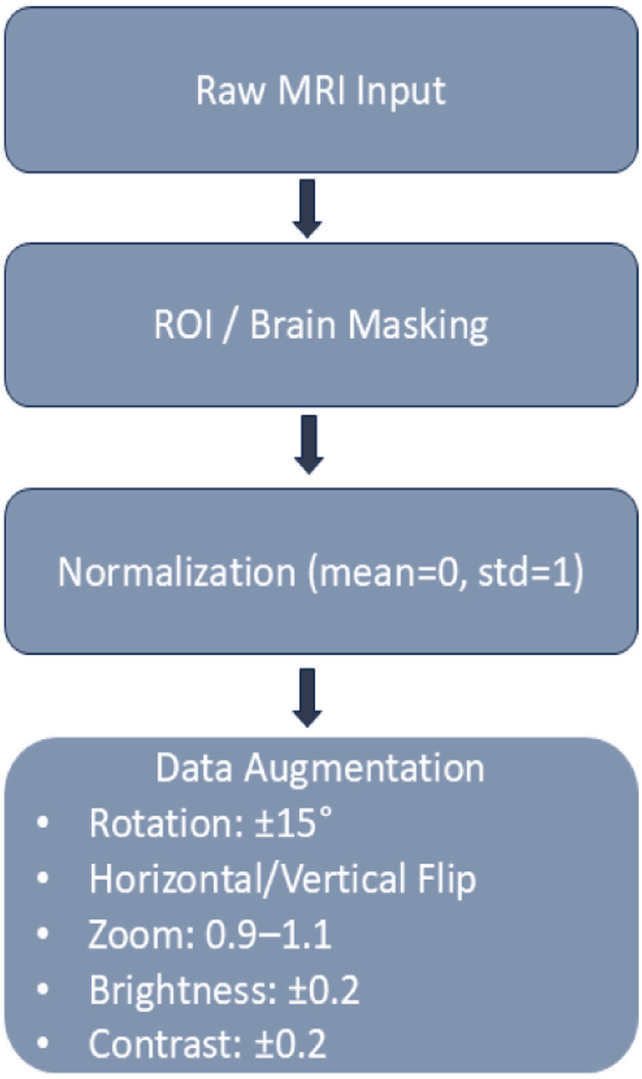


Fig. 16. Structured preprocessing and augmentation pipeline for brain tumor MRI images.

widely adopted as a benchmark dataset in brain tumor classification research. To mitigate potential label noise and dataset bias, duplicate elimination, leakage-aware splitting, and cross-dataset evaluation were conducted. Nevertheless, validation on curated, multi-institutional clinical datasets remains necessary prior to real-world deployment. Accordingly, the findings of this study demonstrate methodological robustness across public datasets rather than definitive clinical reliability.

In a real-world hospital workflow, uncertainty scores can function as triage indicators. Predictions with high epistemic uncertainty can be flagged for secondary radiologist review or additional imaging sequences, thereby reducing risk of misclassification. Conversely, low-uncertainty predictions may accelerate routine case handling. Such selective prediction strategies may optimize diagnostic throughput while maintaining safety thresholds.

5.1. *Clinical translation, deployment, and future validation*

The proposed framework should currently be regarded as a research-stage image-classification method rather than a clinically validated diagnostic system. Translation into clinical practice would require evaluation on independent, institutionally governed MRI datasets representing relevant patient populations, disease spectra, scanners, field strengths, acquisition protocols, and clinical workflows. Particular attention should be paid to differences between two-dimensional images extracted from public repositories and complete clinical MRI examinations, which may include multiple sequences, volumetric information, and additional contextual data.

A deployment pathway should include external multicentre testing, assessment of calibration under distribution shift, and predefined thresholds for referral to expert review. Model outputs should be presented as decision-support information and should not replace radiological interpretation. Uncertainty-based abstention could be investigated as a mechanism for identifying cases requiring additional review, but its safety would need to be established prospectively by examining false-negative cases, referral rates, workload, and subgroup-specific performance.

Technical implementation would need to address DICOM interoperability, image preprocessing consistency, storage and transfer of medical data, cybersecurity, audit trails, software maintenance, and integration with picture archiving and communication systems. Changes in scanner vendors, field strengths, contrast administration, acquisition protocols, or image quality could affect model performance and probability calibration. Any deployment would therefore require ongoing monitoring, version control, drift detection, and a documented process for revalidation after material changes to the imaging environment or

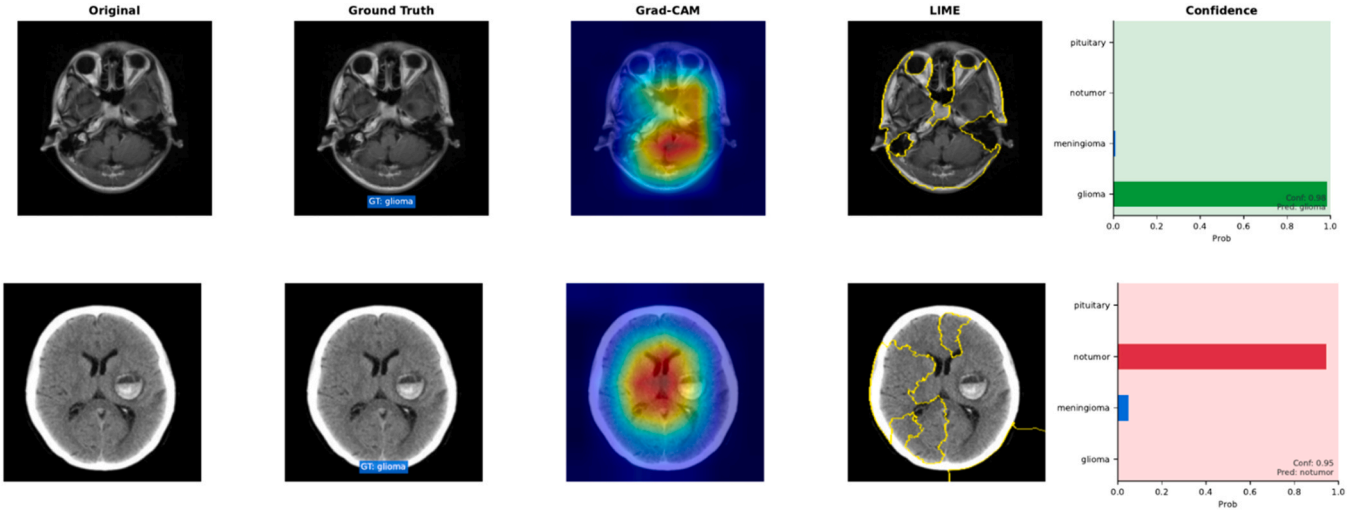


Fig. 17. Explainability analysis using Grad-CAM and LIME for the proposed model on brain tumor classification, showing both correct and incorrect prediction cases.

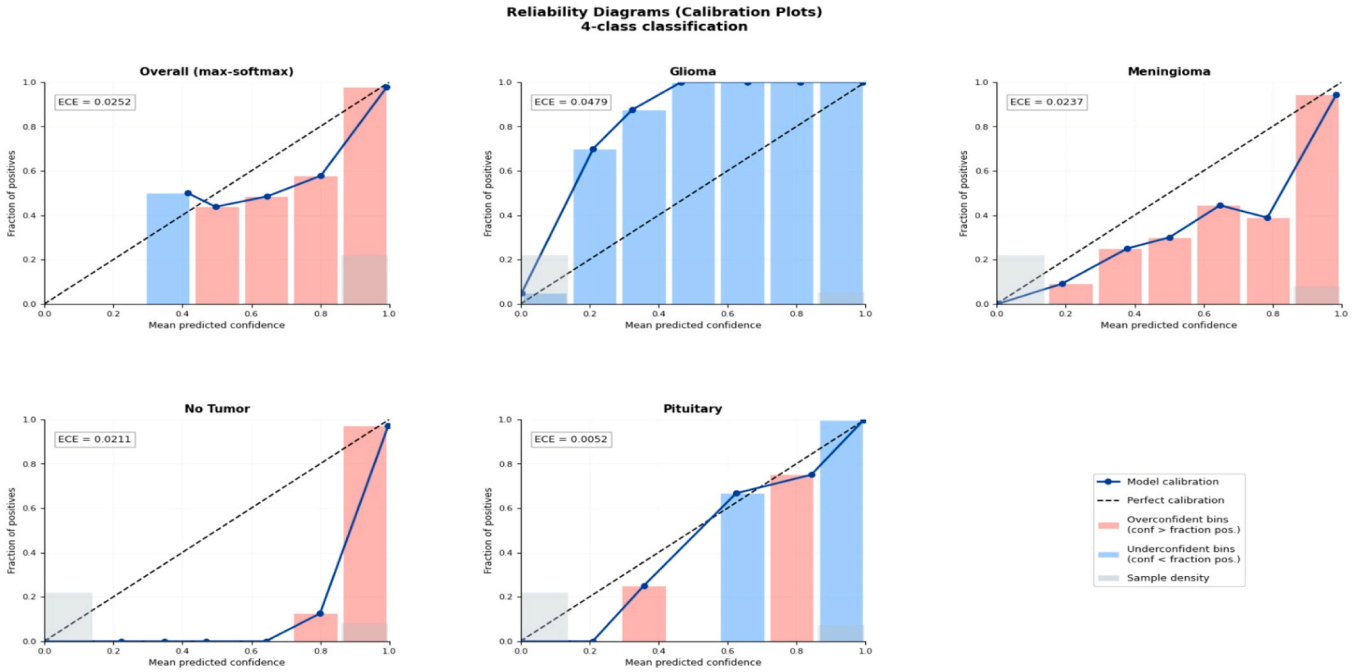


Fig. 18. Reliability diagrams (calibration plots) for Propose model on the brain tumor MRI dataset.

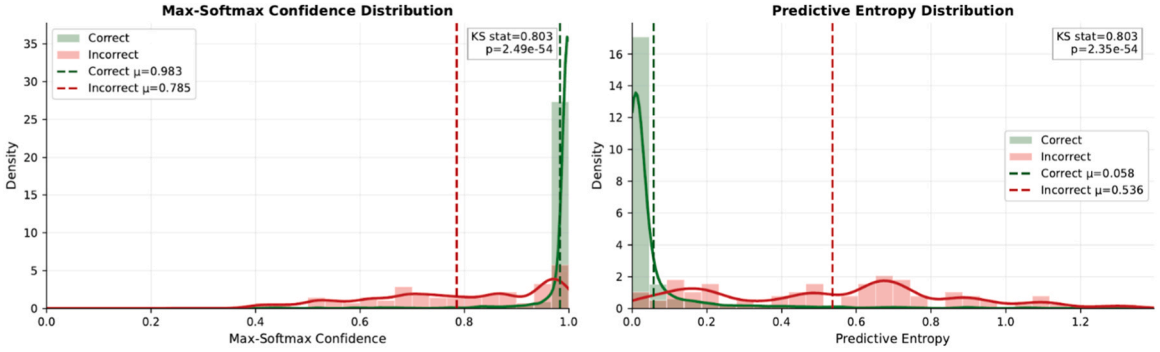


Fig. 19. Uncertainty distributions for correct vs. incorrect predictions.

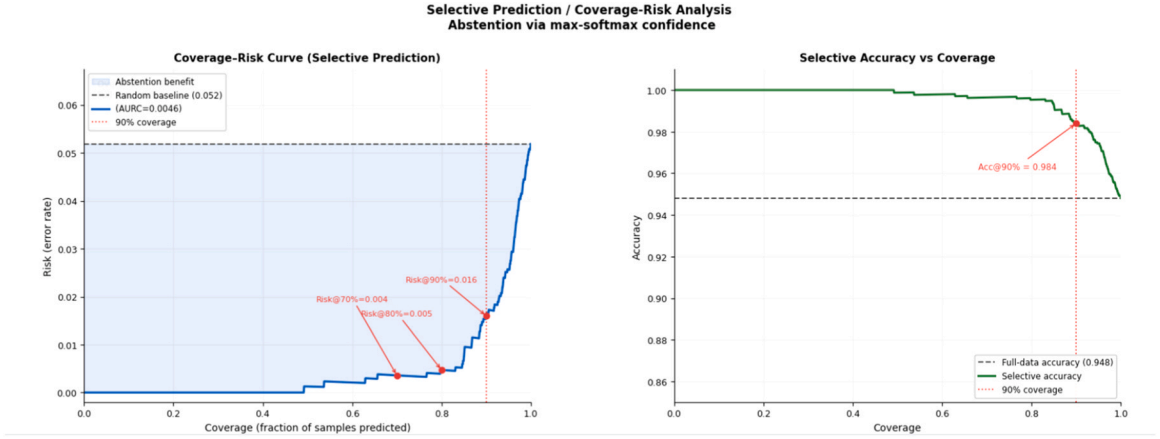


Fig. 20. Selective Prediction and Coverage-Risk Analysis.

model.

Regulatory requirements would depend on the jurisdiction, intended use, clinical claims, and whether the software functions as a medical device. Ethical considerations include patient privacy, informed governance of secondary data use, transparency about model limitations, equitable performance across populations, and accountability for

decisions informed by model output. Explainability visualizations should be evaluated with appropriate reference annotations before they are interpreted as evidence of anatomical localization.

Future research should extend the framework toward multimodal and volumetric MRI, integration of complementary clinical information where justified, and computationally efficient inference suitable for real-world workflows. Federated learning could be explored to support multi-institutional model development without centralizing all patient images, subject to privacy and security safeguards. Longitudinal studies could assess whether the model remains reliable across repeated examinations and changes in disease status or treatment. Ultimately, prospective multicentre evaluation and appropriately designed clinical studies will be needed to establish whether uncertainty-aware classification improves diagnostic workflows, patient outcomes, or resource utilization.

5.2. Limitations

First, the study relies exclusively on publicly available datasets, which may exhibit heterogeneous imaging protocols and annotation variability. Nevertheless, the absence of explicit patient identifiers in the primary Kaggle/Masoud dataset means that similarity-based grouping cannot guarantee complete subject-level separation. The reported primary-dataset performance should therefore be interpreted as performance under a leakage-mitigation protocol rather than definitive evidence of patient-independent generalization. Second, experiments were conducted on 2D MRI slices rather than full 3D volumes, potentially limiting spatial context modeling. Third, while cross-dataset testing was performed, prospective validation on independent clinical cohorts has not yet been conducted. Fourth, although MC Dropout and Deep Ensembles improved calibration, further research is needed to standardize clinically interpretable uncertainty reporting. Finally, the hybrid CNN–Transformer architecture introduces higher computational complexity compared to single-branch models, which may impact deployment feasibility in resource-limited settings.

Overall, the proposed hybrid framework demonstrates consistent performance gains, improved calibration, and qualitative interpretability across public MRI datasets. Future research should therefore progress toward multimodal and volumetric MRI modeling, real-time and resource-efficient inference, federated or other privacy-preserving multi-institutional learning to facilitate collaborative model development without requiring centralized sharing of sensitive imaging data, longitudinal assessment of disease evolution and treatment response, quantitative explainability using expert-annotated localization masks, and prospective clinical trials.

6. Conclusion

This study presents a hybrid CNN–Transformer framework for multiclass brain tumor MRI classification, integrating EfficientNetV2-B0 and Vision Transformer representations through feature-level fusion and an adaptive gating mechanism. The evaluation combines classification performance with probability calibration, predictive uncertainty analysis, ablation experiments, and testing across available public datasets.

The results should be interpreted in the context of the evaluation design and the limitations of the underlying datasets. In particular, the absence of explicit patient identifiers in the primary Kaggle/Masoud dataset prevents confirmation of patient-independent separation, even when exact duplicates and near-duplicate images are screened and similarity-derived groups are separated across partitions. Cross-dataset testing provides additional evidence about performance on the evaluated datasets but does not establish clinical effectiveness.

Future work should prioritize reproducible benchmarking, external multicentre evaluation using documented patient-level partitions, calibration assessment under distribution shift, and prospective studies in clinically representative populations. Until such evidence is available, the proposed framework should be considered a research-stage classification method rather than a validated clinical diagnostic system.

CRediT authorship contribution statement

Tamim Mahmud: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Sanzida Afrin Anamika:** Writing – review & editing, Validation, Methodology, Formal analysis. **Prosenjit Chandra Biswas:** Writing – review & editing, Conceptualization. **Tanvir Mahmud:** Writing – review & editing, Visualization, Formal analysis. **Kulsom Akter Sinthia:** Writing – original draft, Visualization. **Md. Monir Hossain Shimul:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Methodology. **Rahmatul Kabir Rasel Sarker:** Writing – review & editing, Validation, Supervision.

Ethics statement

Not applicable. The study used publicly available, de-identified secondary data.

Funding statement

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets analyzed in this study are publicly available from their respective original sources. The complete analysis code, preprocessing scripts, dataset-splitting procedures, training and evaluation notebooks, uncertainty-estimation routines, ablation scripts, and pretrained model weights are publicly available in the accompanying GitHub repository: https://github.com/tamim1249/Brain-Tumor-Classification. These materials are provided to facilitate independent reproduction of the reported experiments.

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